

# IS THERE A PLATONIC TREE?

## CALIBRATING LATENT HYPERBOLICITY IN FOUNDATION MODELS

**Anonymous authors**

Paper under double-blind review

### ABSTRACT

Hyperbolic methods for representation learning rest on a premise we call latent hyperbolicity: standard models are already tree-like, because their class geometry scores a low Gromov  $\delta$ . We show that this raw reading is confounded by dimension, by covariance spectrum and by the supremum statistic itself. We build an instrument that reads every score as an excess over a random cloud with the same dimension and spectrum, ranks that excess against matched null replicates, and adds a depth test whose power and validity regime are measured. Applied to 12 vision backbones, 6 datasets and 16 text models, the instrument gives a nuanced answer. On image features, where the premise is read, the excess sits within null noise in most cells and is small where it survives. On class centroids, clustered structure is genuine in 49 of 72 cells, but it is rarely deeper than a star. The depth test certifies hierarchy above the superclasses in 4 of 12 ImageNet backbones, and a backbone trained in hyperbolic space shows the same clustering and no additional depth. The trees are moderately shared: a naive comparison isolates the self-supervised family as an island, an artifact of the clustering cut, and once the cut is controlled every training recipe recovers the human taxonomy partially, more so with supervision, with the self-supervised tree living in angles rather than distances. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

## 1 INTRODUCTION

**Is there a Platonic tree?** Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

figures/fig1\_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: **A random cloud, a star and a tree all read as tree-like; only the instrument tells them apart.** Sketched: a high-dimensional random cloud, clusters around a hub and a nested hierarchy, all with a low raw reading, the excess that separates the cloud from the other two, and the depth test that separates the star from the tree.

Semantic knowledge is hierarchical: a husky is a dog, a dog is a mammal, and so on upward. Hyperbolic space is the natural geometric home for such hierarchies (Nickel & Kiela, 2017; Sarkar, 2011). A line of work therefore imposes it on representations at training time, from vision to vision-language and language (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models are said to organize concepts already in a tree-like metric structure, because the Gromov  $\delta$  of their features is low (Khrulkov et al., 2020; Bdeir et al., 2024; Yang et al., 2025; Tifrea et al., 2019). Is the premise right, and what does it license?

The raw reading is confounded. Gromov  $\delta$  assigns zero to trees, but three artifacts push it down without any hierarchy (Gromov, 1987). High-dimensional clouds concentrate toward equidistance, which is a degenerate star tree: we call this the *dimension confound*. Anisotropic spectra mimic low-dimensional behavior: the *spectrum confound*. And the supremum over sampled quadruples is a one-quadruple extreme that does not converge: the *statistic confound*. The three are visible in real models. Small GPT-2 models sit at their matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the *instrument* that reads  $\delta$  correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose power and validity regime are measured. We apply it at the sample level, where the premise is read, and at the class level, where hierarchy lives. With it we ask of geometry what convergence work asks of similarity: what survives calibration, and is it shared, is there a Platonic tree (Huh et al., 2024; Gröger et al., 2026)? They calibrate similarity across models; we calibrate geometry within a model, and the comparison of trees across models turns out to need a calibration of its own.

**The answer has three parts.** On image features the excess sits within null noise in most cells and is small where it survives, so the premise does not survive calibration as stated. On class centroids, clustered structure is genuine in 49 of 72 cells, but it is rarely deeper than a star. The depth test certifies hierarchy above the superclasses in 4 of 12 ImageNet backbones, and a backbone trained in hyperbolic space shows the same clustering and no additional depth. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled every recipe recovers the human taxonomy partially, more so with supervision, with the self-supervised structure living in angles. What follows for practice is what to measure before imposing curvature and which zero-cost readout collects what is there. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

#### Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured power; the raw reading misleads at the sample level and at the class level alike.
2. **The census.** Clustered structure in most models, hierarchy certified in a few, and a hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

The premise and its confounds have a history, which we review before building the instrument.

## 2 RELATED WORK AND BACKGROUND

**The premise.** Hyperbolic representation learning builds on tree embeddability and Poincaré embeddings and now spans vision, vision-language, graphs and language (Sarkar, 2011; Nickel & Kiela, 2017; Ganea et al., 2018; Chami et al., 2019). Its stated premise rests on a low raw reading for CNN and ViT features, for token embeddings and for word vectors (Khrulkov et al., 2020; Bdeir

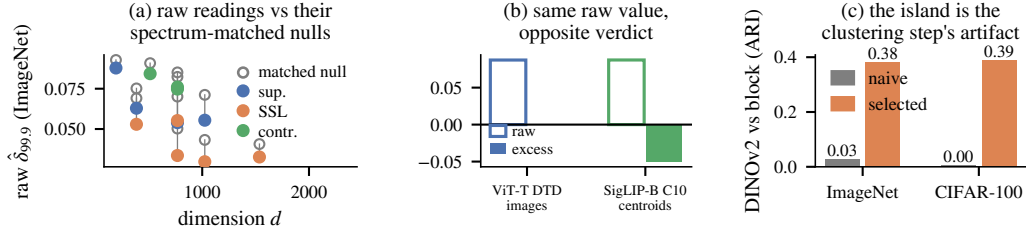


Figure 2: **The raw reading cannot be read on its own, and the naive tree comparison manufactures the island.** Plotted: (a) the raw reading of the 12 backbones on ImageNet against their matched null means, shown hollow; (b) a sample-level cell (ViT-T on DTD images) and a class-level cell (SigLIP-B on CIFAR-10 centroids) with the same raw reading and opposite verdicts; (c) DINOv2-vs-block tree agreement before and after the configuration control of Section 5.

et al., 2024; Yang et al., 2025; Tifrea et al., 2019). Those readings are sample-level, uncalibrated and taken under a single construction of the statistic, and Section 4 re-reads them with the instrument.

**Calibrating geometry against nulls.** Network science measures Gromov  $\delta$  on graphs against random graphs of matched size and degree sequence, and found long ago that the raw reading does not certify tree-likeness (Narayan & Saniee, 2011; Kennedy et al., 2013; Adcock et al., 2013). The confound itself is therefore not our contribution. Representation similarity, intrinsic dimension and neural collapse characterize Euclidean structure without testing tree-likeness, and neural collapse is the flat limit we compare our reading with in Section 4 (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappan et al., 2020). The Platonic Representation Hypothesis argues that models converge toward shared representations; Koepke et al. (2026) stress-test that convergence across modalities, Gröger et al. (2026) show that permutation-calibrated similarity removes most of it while local structure survives, and Park et al. (2024) read concept hierarchy in language models from orthogonality (Huh et al., 2024). What is new here is the instrument for embedding clouds rather than graphs: spectrum-matched nulls at the resolution of the null replicate, a depth test against matched stars with measured power, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

**Background.** For four points  $w, x, y, z$  of a metric space, sort the three pairing sums  $d(w, x) + d(y, z)$ ,  $d(w, y) + d(x, z)$ ,  $d(w, z) + d(x, y)$  as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ ; the four-point defect is  $(S_{\max} - S_{\text{mid}})/2$  and  $\delta$  is its supremum. Trees have  $\delta = 0$ , and under this definition the hyperbolic plane has  $\delta = \ln 2 \approx 0.693$ , distinct from the thin-triangles constant  $\ln(1 + \sqrt{2}) \approx 0.881$  (Nica & Špakula, 2016). We use the normalized  $\delta_{\text{norm}} = \delta_{\max}/\text{diam}$  throughout: hyperbolic regions drift to zero as they grow while spheres stay high, so the ordering trees < hyperbolic < spherical holds. Sampled-quadruple estimation has known bias and scale caveats, which the calibration-plus-null design addresses operationally, and Ollivier–Ricci curvature on a kNN graph is the local companion, positive within clusters and negative on bridges (Fournier et al., 2015; Ollivier, 2009). How the raw reading becomes a calibrated one is the subject of the next section.

### 3 THE INSTRUMENT

If the premise were right, a low raw reading would be evidence on its own. The dimension confound says it is not, so this section builds the measurement layer in whose units every later claim is stated.

**The premise lives at the sample level; hierarchy lives at the class level.** Each frozen model is a fixed feature extractor. The premise is read on per-image features, which is where the literature measures it. Hierarchy is measured on class centroids: in vision, the mean of each class’s cached training embeddings, a hundred per class on ImageNet; in text, the embeddings of “a photo of a {class}” over the ImageNet classes. A *cell* is one pair of model and class set, and the census covers 12 vision backbones on 6 datasets and 16 text models. The raw reading is the 99.9th-percentile four-point statistic  $\hat{\delta}_{99.9}$  over  $5 \times 10^5$  sampled quadruples per seed, averaged over 10 seeds; the

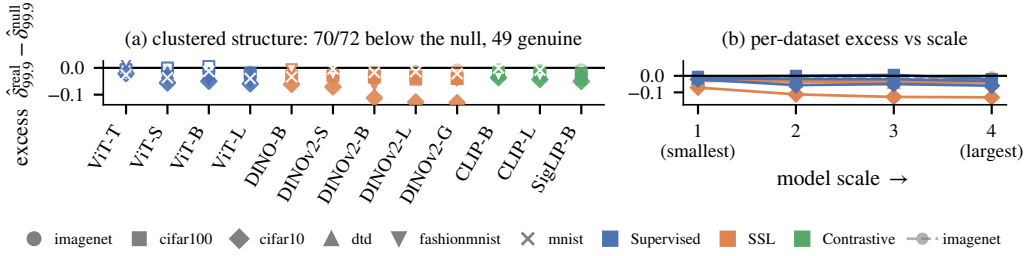


Figure 3: **Clustered structure is the rule at the class level.** Plotted: (a) the excess of every cell of the census, with hollow markers for cells that are not genuine; (b) the excess against model scale for each dataset, where only DINOv2 on CIFAR-10 deepens monotonically.

supremum, the pre-registered statistic, is reported in the appendix for the reason given below. The excess is budget-stable, and on ImageNet every sign holds at every budget (Appendix A.15).

**A low raw reading is not evidence.** We calibrate the estimator on reference geometries with known answers. It recovers the tree zero and the  $\ln 2$  four-point constant of  $\mathbb{H}^2$ , whose normalized reading drifts to zero as the region grows. Spheres stay high. Iid Gaussians, which have no structure at all, drift lower as the dimension grows, from 0.104 at  $d=192$  to 0.046 at  $d=1536$ . That drift is the dimension confound in its purest form: the raw reading of a real backbone must be compared with the reading of a structureless cloud of the same shape (Table 3). The next paragraph builds that comparison.

**The reading is an excess over a matched null.** For every cell we synthesize null clouds with the same  $n$ ,  $d$ , mean and covariance spectrum. We use the Haar construction throughout: random orthogonal coefficients recombined with the real singular values, so the sample spectrum is exact; the Gaussian-coefficient construction is the alternative reported in the appendix. The **excess** is the raw reading minus the mean reading of 200 such replicates. The primary evidence is the *percentile rank*, the number of replicates above the real value, and its left-tail  $p$ . A cell is *genuine* when its Benjamini–Hochberg-corrected  $p$  across the census is at most 0.05; we use the word in this sense only. A negative genuine excess certifies *clustered structure*; the word “tree-like” is reserved for the depth test of Section 4. A positive excess means less tree-like than chance given the spectrum (Appendix A.1).

**Two choices changed after pre-registration.** We disclose both. The supremum statistic did not converge under the quadruple budget on ImageNet and disagreed with the 99.9th percentile of the same defects in both directions; the percentile is therefore the raw reading and the supremum is reported in full in the appendix. The Gaussian-coefficient null does not reproduce the sample spectrum it claims to match when  $n$  is small relative to  $d$ , and it inflates the excess of low-rank clouds; the Haar null does reproduce it, so we use the Haar null and report both. Rank-based calibration is what gives permutation scores their finite-sample validity, so a  $z$ -score on a handful of replicates is kept only as a secondary summary (Gröger et al., 2026, §5.1). The null is aggregation-aware in the sense of Gröger et al. (2026, §5.2): the same percentile of the same sampled defects is evaluated on every replicate, so the max-statistic inflation is calibrated rather than ignored (Table 31).

**Projection and tasks.** A parameter-free map centers the features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$  and preserves directions, so Euclidean, Poincaré and cosine readouts can be compared on the intact representation; nearest-centroid and 5-way 5-shot classification score them for Section 6 (Appendix A.10). With the instrument in place, we read the premise where it is read.

| model    | family | class level (centroids)  |                |                 |                 |                |             | sample level (images) |                 |
|----------|--------|--------------------------|----------------|-----------------|-----------------|----------------|-------------|-----------------------|-----------------|
|          |        | $\hat{\delta}_{99.9}$ IN | excess IN      | $r/200$ ( $p$ ) | exc. C100       | exc. C10       | $\rho_{WN}$ | exc. C100             | exc. DTD        |
| ViT-T    | sup.   | 0.088                    | $-0.005^\circ$ | 169 (.159)      | $-0.006^\circ$  | $-0.021^\circ$ | +0.52       | $-0.002^\circ$        | $-0.000^\circ$  |
| ViT-S    | sup.   | 0.063                    | $-0.012$       | 200 (.005)      | $-0.000^\circ$  | $-0.056$       | +0.50       | +0.005 $^\circ$       | $-0.006$        |
| ViT-B    | sup.   | 0.054                    | $-0.016$       | 200 (.005)      | +0.006 $^\circ$ | $-0.050$       | +0.49       | +0.003 $^\circ$       | $-0.008$        |
| ViT-L    | sup.   | 0.055                    | $-0.016$       | 200 (.005)      | $-0.020$        | $-0.060$       | +0.53       | $-0.004^\circ$        | $-0.007$        |
| DINO-B   | SSL    | 0.055                    | $-0.020$       | 200 (.005)      | $-0.032$        | $-0.061$       | +0.36       | $-0.004^\circ$        | +0.001 $^\circ$ |
| DINOv2-S | SSL    | 0.053                    | $-0.016$       | 200 (.005)      | $-0.031$        | $-0.071$       | +0.22       | $-0.005$              | +0.006 $^\circ$ |
| DINOv2-B | SSL    | 0.034                    | $-0.017$       | 200 (.005)      | $-0.031$        | $-0.112$       | +0.18       | $-0.009$              | $-0.002^\circ$  |
| DINOv2-L | SSL    | 0.030                    | $-0.014$       | 200 (.005)      | $-0.035$        | $-0.128$       | +0.18       | $-0.013$              | $-0.006$        |
| DINOv2-G | SSL    | 0.033                    | $-0.008$       | 200 (.005)      | $-0.040$        | $-0.130$       | +0.20       | $-0.023$              | $-0.008$        |
| CLIP-B   | contr. | 0.084                    | $-0.007^\circ$ | 172 (.144)      | $-0.019$        | $-0.036$       | +0.57       | $-0.006^\circ$        | +0.002 $^\circ$ |
| CLIP-L   | contr. | 0.076                    | $-0.009$       | 197 (.020)      | $-0.018$        | $-0.043$       | +0.59       | +0.004 $^\circ$       | $-0.001^\circ$  |
| SigLIP-B | contr. | 0.075                    | $-0.008^\circ$ | 189 (.060)      | $-0.022$        | $-0.050$       | +0.58       | $-0.007$              | $-0.004^\circ$  |

Table 1: **Clustered structure is the rule at the class level; the sample-level reading sits within null noise in most cells.** One row per backbone (supervised / self-supervised / contrastive blocks).  $\hat{\delta}_{99.9}$ : raw ImageNet reading of the 99.9th-percentile four-point statistic; excess:  $\hat{\delta}_{99.9}$  minus the mean of 200 Haar-rotated spectrum-matched null replicates;  $r/200$ : replicates above the real value, with the left-tail  $p = (1 + \#\{\text{null} \leq \text{real}\})/201$ ; a cell is *genuine* when its Benjamini–Hochberg-corrected  $p \leq 0.05$  over the 72 cells;  $^\circ$ : not genuine; bold: sign-positive, i.e. less clustered than the null.  $\rho_{WN}$ : Spearman correlation of inter-centroid and WordNet distances. Sample level: the same reading on  $\approx 1000$  stratified training images per cell, BH over those 24 cells: genuine in 10 of 24 (full table in Appendix A.18). Cross-dataset magnitudes are not comparable (§4). DTD, the flat datasets, the supremum reading and the task columns are in Appendix A (Table 5); all four null  $\times$  statistic verdicts per cell in Table 31.

#### 4 FINDINGS I: LATENT HYPERBOLICITY, CALIBRATED

If standard models were latently hyperbolic, the instrument should find an excess on their image features, where the premise is read. Does it, and does anything survive on class centroids, where hierarchy lives?

**The premise does not survive calibration.** We read it first on per-image features, on roughly a thousand stratified training images per cell for CIFAR-100 and DTD and all twelve backbones. The raw reading lies in the low band the literature reports. Calibrated, it is not genuine in 14 of 24 cells, and where it is genuine the excess is small and concentrated in the self-supervised family. A low raw reading on image features is therefore what a random cloud of the same dimension and spectrum produces, up to a small margin. This is the reading that motivates hyperbolic methods, and it does not survive calibration as stated (Table 1, last two columns). Class centroids, where hierarchy lives, read differently, and we turn to them next.

**Clustered structure is the rule.** Figure 3 and Table 1 give the census of record. Almost every cell sits below its matched null, and in 49 of the 72 the gap is one that chance does not produce. On ImageNet and CIFAR-100, the two sets with a real hierarchy, the count is 18 of 24. The structure belongs to no single family or architecture: supervised, self-supervised and contrastive backbones all contribute genuine cells, two self-supervised ResNets read the same as the ViTs, and resampling the images behind each centroid does not move the reading. What does move it is the label set. The smallest backbone is genuine on one dataset only, the flat datasets sit at the resolution limit of ten classes, and a class-count control shows that the reading follows the hierarchy of the labels rather than their number (Appendix A.8). A star, however, would pass this test too, which is the subject of the next paragraph.

**A star passes the census.** The excess measures structure beyond second moments, and a pure star already has such structure. A synthetic star of thirty clusters with no hierarchy scores an excess as deep as any real cell. We call this the *star caveat*. The excess therefore measures how classes are

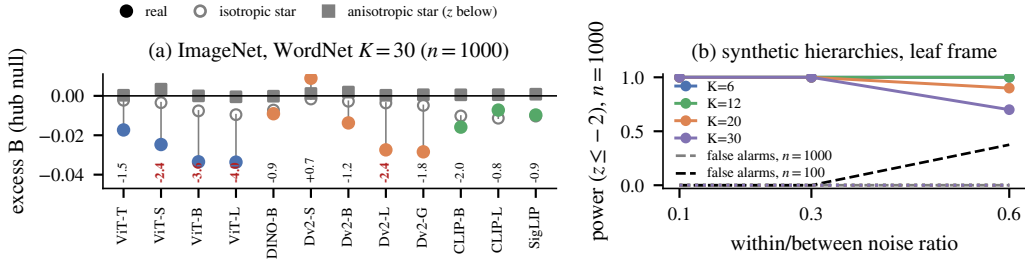


Figure 4: **The depth test certifies hierarchy above the superclasses in 4 of 12 ImageNet backbones.** Plotted: (a) the excess under the hub-randomizing null for the real ImageNet centroids and for their matched stars, isotropic as hollow circles and anisotropic as squares, with the  $z$  of real minus anisotropic star under each backbone, red at  $z \leq -2$ ; (b) the power of the test on synthetic hierarchies at ImageNet’s class count, one line per  $K$ , with the false-alarm rates of pure stars dashed.

arranged, not the geometry they are embedded in, and a negative excess says clustered, not deep. Whether the arrangement is deeper than a star is a separate question, which needs a test of its own (Appendix A.5). That test is the subject of the next paragraph.

**Depth is certified on ImageNet and not on CIFAR-100.** The depth test compares the real centroids with a matched star: Gaussian hubs at the real hub radius and, within each cluster, a Haar sample with the cluster’s own covariance, both read under a null that randomizes the hubs and keeps every cluster intact. Because the null keeps the clusters, the test certifies hierarchy above the labelled clusters only. We measured its power on synthetic hierarchies against a bar fixed before the sweep. The bar fails for class sets of a hundred, where clusters hold fewer than ten points and pure stars are declared hierarchical in 17% of runs, and it is met for class sets of a thousand, where no star is ever declared hierarchical. We therefore certify the test at ImageNet’s class count only, a scoping adopted after the sweep. There, the real cloud is more hierarchical than its matched star in 4 of 12 backbones, three supervised ViTs and DINOv2-L, and never less. CIFAR-100 lies in the unvalidated regime and its readings stay unvalidated. An earlier isotropic star, blind to each cluster’s covariance, produced false alarms and is withdrawn, and the hierarchically fine-tuned checkpoints that would serve as a positive control were not stored (Figure 4). Depth is therefore rare, and whether imposing the geometry creates it is the next question.

**Imposing the geometry does not create depth.** MERU embeds images and text on the Lorentz hyperboloid with an entailment objective, and its released Euclidean twin is a CLIP baseline with the same backbone, data and recipe minus the hyperbolic lift, so the pair isolates the geometry (Desai et al., 2023). Passed through the instrument unchanged, the two read the same excess on ImageNet images, and MERU’s reading in its own Lorentz metric equals its Euclidean reading to the fourth decimal. Under the depth test, in the regime where it is validated, neither model is more hierarchical than a matched star, and MERU is never ahead of its twin. Training in hyperbolic space therefore neither creates the clustering nor deepens it (Table 10). If the geometry does not explain the clustering, the training objective might, and neural collapse is the first candidate.

**Neural collapse is the flat limit, not the reading.** Neural collapse predicts that the class means of a supervised classifier converge to a simplex equiangular tight frame (Papayan et al., 2020). An exact frame is the extreme case of the star caveat: every four-point defect vanishes and  $\delta$  is zero with no hierarchy at all. Our reading differs from it in two ways. The arrangement we certify is not flat, because class means cluster around superclass hubs that a frame would erase, and cutting the centroid dendrograms recovers the CIFAR-100 superclasses. The same clustered structure also appears in models trained without class labels, on transfer datasets and in causal language models, outside the setting where neural collapse is stated. The depth test then asks what neural collapse does not, whether there is hierarchy above the hubs, and answers yes for four ImageNet backbones, three of them supervised ViTs (Appendix A.16). What moves the excess, if not the objective’s flat limit, is scale and training, which we take up last.

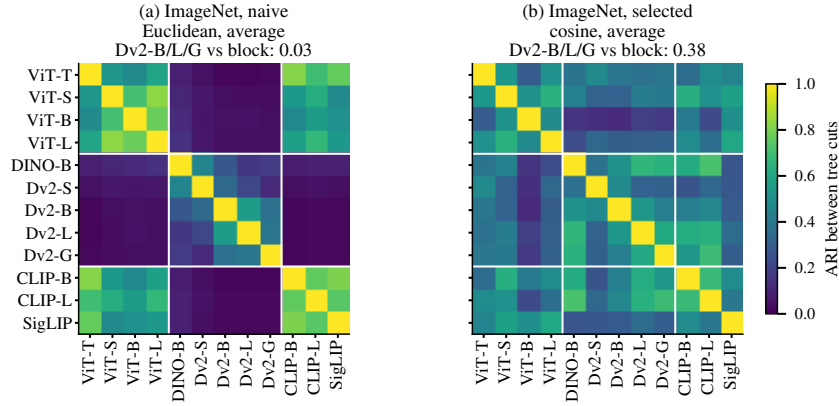


Figure 5: **The island vanishes once the clustering step is controlled.** Plotted: pairwise ARI between the dendrogram cuts of the 12 vision models on ImageNet under the naive configuration (a) and the selected one (b), with the mean DINOv2-vs-block ARI above each panel.

**Scale deepens the excess for one family only.** Across the census, model scale does not deepen the excess in general. Figure 3b shows the one exception: DINOv2, whose excess deepens with size on CIFAR-10 and, weakly, on CIFAR-100. Training moves the raw reading within an architecture. Fine-tuning on a task with no class hierarchy raises it, it falls across transformer depth, and shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should (Appendix A.6). Together, these results say that what survives at the class level is clustered structure that is genuine in most cells, rarely deep, and not created by the geometry of the embedding space. Whether that structure is shared across models, and whose tree it is, is the subject of the next section.

## 5 FINDINGS II: WHOSE TREE

If the measured trees were a shared representation of reality, models should agree with each other and with human taxonomy. Whether they appear to depends on how the trees are measured.

**The naive map manufactures an island.** We extract each model’s hierarchy as a dendrogram over the shared centroids and compare hierarchies rather than representations, by the adjusted Rand index between cut partitions and by the cophenetic correlation between tree and distances. Under the naive configuration, Euclidean distances with average linkage, the map is spectacular: supervised ViTs and contrastive VLMs form one block, and the label-free DINOv2 family appears as an island that agrees with no supervised model. We call this the *island*. Its cuts are degenerate: DINOv2’s Euclidean dendrograms chain, almost every class falls into one cluster, and the index is forced to zero by construction. A configuration must therefore be selected by a criterion fixed without reference to the agreements being compared: we require the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model, which selects cosine-average on ImageNet and cosine-complete on CIFAR-100. Figure 5 shows the map before and after this control (Appendix A.9). Under the selected configurations the island has to be re-read.

**The island is an artifact, and sharing is graded.** Three measures give the same reading under the selected configurations. At the cut, the island shrinks from categorical to moderate: DINOv2 agrees with the block at 0.38 against 0.48 within the block. In cophenetic correlation a gap remains. In triplet agreement, which needs no cut, the gap closes entirely under angular configurations and persists under Euclidean ones. Across every non-degenerate configuration, label-free models recover the human taxonomy partially and supervision deepens the resemblance: paired within each admissible configuration, the label-supervised ViT recovers more than each DINOv2 model in 11 of 12 comparisons (Appendix A.9). The gap is a property of the measure as much as of the models, and its persistence under Euclidean measures alone points to where the self-supervised structure lives.

**The self-supervised tree lives in angles.** DINOv2’s semantics concentrate in the angular component of its representation: for DINOv2-L, sibling triplets are resolved far better under cosine than under Euclidean distance. We call this structure the *angular tree*. It is why DINOv2’s Euclidean dendrograms chain into the island, and why the triplet gap closes only under angular configurations. The census agrees when we repeat it on cosine geometry, with L2-normalized centroids, geodesic distances and the same Haar null on the normalized cloud: the verdict matches the Euclidean census in 65 of 72 cells, and DINOv2 on ImageNet is genuine under both readings. We report the cosine census as robustness, never as the record (Table 33). Where the structure lives decides which readout collects it, which is the concern of Section 6. First we ask whose taxonomy the trees recover.

**WordNet alignment follows supervision, not tree-likeness.** The correlation between inter-centroid distances and WordNet tree distances on ImageNet orders the families by supervision: highest for the contrastive VLMs, then the supervised ViTs, and lowest for DINOv2, with shuffle and Gaussian controls at zero. DINOv2 is nevertheless genuine on ImageNet and on both CIFAR sets. Cutting centroid dendrograms recovers the true CIFAR-100 superclasses at an adjusted Rand index of up to 0.61, against zero for permuted labels. Across the three families, WordNet content and robustness of the excess anti-correlate: the supervised and contrastive families are the most aligned and the most fragile, DINOv2 the least aligned and the most robust (Table 1). Alignment with the human taxonomy is therefore something training puts in, not something tree-likeness reveals, which raises the question of circularity on ImageNet.

**The recovery is not WordNet circularity.** ImageNet classes are WordNet synsets, so recovering WordNet there could be circular. Three controls bound the circularity. WordNet-coherent groupings read a lower raw  $\delta$  than random ones in 11 of 12 models. Single-image distances still correlate with WordNet, up to a correlation of +0.26. And the decoupling of alignment from robustness is the opposite of what label injection would produce. Repeating the tree map on DBpedia Classes, a three-level ontology independent of WordNet, closes the door: no configuration degenerates, the three text embedders recover the true level-2 partition in every configuration, and they agree with each other (Table 39). The recovered hierarchy is real, shared and configuration-robust, so the fragility above belongs to the family and the concept set together, which text makes explicit.

**In text, clustered structure depends on recipe, scale and probe.** On DBpedia, a real three-level text hierarchy, the sentence embedders read a genuine excess with every replicate above the real value. On the harder probe, the thousand ImageNet class names, the same embedders are not genuine, despite raw readings indistinguishable from vision backbones whose excess is. Presence of structure is thus a property of the model and the concept set together. Within causal language models on the class-name probe, the two largest GPT-2 models are genuine under every construction of the null and statistic and under cosine, whereas the two smallest flip across readings: robust at the larger scales, fragile at the smaller. Pythia is genuine at every scale tested, and so is OLMo-1B. Extraction protocol and prompt template move the fragile values but not the scale ordering (Appendix A.7). The trees, then, are shared in content only partially, and whether they are shared in local structure at all is the last question of this section.

**Local agreement survives calibration.** We apply the permutation calibration of Gröger et al. (2026) to the 66 model pairs: permutations of the class correspondence give the null distribution, its upper quantile gives the threshold, and the calibrated score is the excess over that threshold rescaled to keep the maximum at one. Mutual-kNN agreement survives calibration with every pair significant, and so does centroid CKA. The Poincaré reading improves neither. Centroids therefore distill a relational base that models share locally, in who is near whom, even where their trees differ (Table 22). Together, these findings give the thesis its full form. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

## 6 CONSEQUENCES FOR IMPOSING CURVATURE

If the raw reading cannot be read on its own, what should a practitioner read before imposing curvature? And if the structure that survives is clustered and angular, which readout collects it?

**The raw reading cannot select a curvature.** Khruikov et al. (2020) set the curvature of the Poincaré ball from the measured relative hyperbolicity as  $c = (0.144/\delta_{\text{rel}})^2$  with  $\delta_{\text{rel}} = 2\delta/\text{diam}$ , the constant being the effective value of the ball at their clipping radius. Since the raw reading is dominated by the dimension and spectrum confounds, a curvature chosen this way tracks the backbone’s width rather than its hierarchy. The iid Gaussian clouds of our calibration, which have no structure at all, would receive a curvature from 0.48 at  $d=192$  to 2.5 at  $d=1536$ . The excess is the quantity to read instead, and it is small: it runs from  $-0.005$  to  $-0.020$  on ImageNet centroids and sits within noise in most cells on image features (Table 1). A practitioner who cannot read curvature from the raw reading needs to know what to read instead, which is the next paragraph.

**Measure before imposing.** For a frozen backbone the instrument answers three questions in minutes. Is there structure beyond a spectrum-matched null: the excess and its rank. Is that structure deeper than a star: the depth test, in the regime where it is validated. And where does it live, in angles or in distances: the cosine reading of Section 5. In this census the answers are usually yes, rarely, and, for the self-supervised family, in angles. What remains is to collect the structure that is there, which a change of readout does at no cost.

**A zero-cost readout collects what is there.** Changing the readout metric on the intact representation costs nothing. Cosine collects the angular tree. The Poincaré readout adds from  $+0.1$  to  $+1.3$  pp over cosine for the contrastive VLMs on prototype tasks. For the other families its effect is inconsistent, from  $-1.7$  to  $+1.1$  pp. On flat-label datasets no metric helps, and sample-level tasks never benefit. The full study, with its confidence intervals and its negatives, is in the appendix (Appendix A.10). These consequences close the three acts, which the discussion states together.

## 7 DISCUSSION AND LIMITATIONS

**What the paper establishes.** Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry. The thesis has three acts. The premise of latent hyperbolicity is confounded, and its sample-level reading sits within null noise in most cells. What survives at the class level is clustered structure that is genuine in 49 of 72 cells and in 18 of 24 on ImageNet and CIFAR-100, hierarchy above the superclasses in 4 of 12 ImageNet backbones, no additional depth from imposing the geometry, and a moderately shared, supervision-dependent recovery of the human taxonomy whose self-supervised part is the angular tree. What follows is that curvature should be measured rather than assumed, and that a zero-cost readout collects what is there.

**Open questions.** Why the structure is clustered but rarely deep is open. Every recipe recovers the superclass hubs, yet only four backbones place them beyond a star. Why the self-supervised tree is angular is the second question, and the objective’s invariances are the natural suspect.

**Limitations.** (i) The excess certifies structure beyond second moments, star-like clustering included; the depth test certifies hierarchy above its frame only, is validated for class sets of ImageNet’s size, and does not test hierarchy within the labelled clusters (Appendix A.5). (ii) The census is class-centroid geometry on the datasets of Table 1; other concept sets may expose different structure. (iii) Causal-LM values, not their scale ordering, are template-conditioned. (iv) The objective–geometry link is correlational and triplet-bounded, and the within-dataset prediction of the metric gain rests chiefly on DINOv2’s scale range, as Appendix A.13 shows. (v) Downstream scope is prototype-based metric tasks with modest gains. (vi) The parallelogram descriptor  $\xi$  tracks norm dispersion rather than curvature, and Appendix A.21 reports it only as such. (vii) A causal LM’s class representation depends on the extraction protocol as much as on the model: left-padded batching moves the raw reading by an amount of the order of the fragile excesses. (viii) The sample-level reading covers two datasets and one image budget.

## ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic.

We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

## REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it; all tables are generated by script from those files, and the full pipeline (feature extraction, diagnostics, nulls, calibration, tasks, statistics, figures) with fixed seeds will be released. Estimator settings are stated in §3. The instrument ships as one script (ANONYMIZED-REPO) that reproduces any cell of Table 1 from its centroid matrix; its output on two cells is checked against the table at every freeze. Computing  $\delta_{\text{norm}}$  on 1000 centroids takes seconds on CPU; the full task grid runs in under 30 minutes on one consumer GPU.

## THE USE OF LARGE LANGUAGE MODELS

LLM-based assistants were used, under continuous author direction and review, to draft and edit text, to write analysis scripts, and to run the experiment pipeline on the authors’ hardware. All experimental designs, results, verifications, and final claims were checked and approved by the authors, who take full responsibility for the content of this paper.

## REFERENCES

- Aaron B. Adcock, Blair D. Sullivan, and Michael W. Mahoney. Tree-like structure in large social and information networks. In *IEEE International Conference on Data Mining (ICDM)*, pp. 1–10, 2013.
- Morris Alper and Hadar Averbuch-Elor. Emergent visual-semantic hierarchies in image-text representations. In *European Conference on Computer Vision (ECCV)*, 2024.
- Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 6109–6119, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/cfcce0621b49c983991lead4c3d4d3b6b-Abstract.html>.
- Mina Ghadimi Atigh, Julian Schoep, Erman Acar, Nanne van Noord, and Pascal Mettes. Hyperbolic image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4453–4462, June 2022.
- Ahmad Bdeir, Kristian Schwethelm, and Niels Landwehr. Fully hyperbolic convolutional neural networks for computer vision. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=ekz1hN5QNh>.
- Ines Chami, Zhitao Ying, Christopher Ré, and Jure Leskovec. Hyperbolic graph convolutional neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 4869–4880, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/0415740eaa4d9decbcb8da001d3fd805f-Abstract.html>.
- Weize Chen, Xu Han, Yankai Lin, Hexu Zhao, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. Fully hyperbolic neural networks. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5672–5686, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.389. URL <https://aclanthology.org/2022.acl-long.389/>.

- Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, and Ramakrishna Vedantam. Hyperbolic image-text representations. In *Proceedings of the 40th International Conference on Machine Learning, ICML'23*. JMLR.org, 2023.
- Aleksandr Ermolov, Leyla Mirvakhabova, Valentin Khrulkov, Nicu Sebe, and Ivan V. Oseledets. Hyperbolic vision transformers: Combining improvements in metric learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pp. 7399–7409. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00726. URL <https://doi.org/10.1109/CVPR52688.2022.00726>.
- Hervé Fournier, Anas Ismail, and Antoine Vigneron. Computing the gromov hyperbolicity of a discrete metric space. *Information Processing Letters*, 115(6-8), 2015.
- Octavian Ganea, Gary Becigneul, and Thomas Hofmann. Hyperbolic neural networks. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL [https://proceedings.neurips.cc/paper\\_files/paper/2018/file/dbab2adc8f9d078009ee3fa810bea142-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2018/file/dbab2adc8f9d078009ee3fa810bea142-Paper.pdf).
- Fabian Gröger, Song Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An aristotelian view. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.14486.
- M. Gromov. *Hyperbolic Groups*, pp. 75–263. Springer New York, New York, NY, 1987. ISBN 978-1-4613-9586-7. doi: 10.1007/978-1-4613-9586-7\_3. URL [https://doi.org/10.1007/978-1-4613-9586-7\\_3](https://doi.org/10.1007/978-1-4613-9586-7_3).
- Neil He, Rishabh Anand, Hiren Madhu, Ali Maatouk, Smita Krishnaswamy, Leandros Tassiulas, Menglin Yang, and Rex Ying. HELM: Hyperbolic large language models via mixture-of-curvature experts. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=RnbJPkakkM>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, Proceedings of Machine Learning Research, pp. 20617–20642. PMLR / OpenReview.net, 2024. URL <https://proceedings.mlr.press/v235/huh24a.html>.
- W. Sean Kennedy, Onuttom Narayan, and Iraj Saniee. On the hyperbolicity of large-scale networks. *arXiv preprint arXiv:1307.0031*, 2013.
- Valentin Khrulkov, Leyla Mirvakhabova, Evgeniya Ustinova, Ivan Oseledets, and Victor Lempitsky. Hyperbolic image embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- A. Sophia Koepke, Daniil Zverev, Shiry Ginosar, and Alexei A. Efros. Back into plato’s cave: Examining cross-modal representational convergence at scale, 2026. arXiv preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey E. Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov (eds.), *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, Proceedings of Machine Learning Research, pp. 3519–3529. PMLR, 2019. URL <http://proceedings.mlr.press/v97/kornblith19a.html>.
- Onuttom Narayan and Iraj Saniee. Large-scale curvature of networks. *Physical Review E*, 84(6): 066108, 2011.
- Bogdan Nica and Ján Špakula. Strong hyperbolicity. *Groups, Geometry, and Dynamics*, 10(3), 2016. arXiv:1408.0250.

- Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. In Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pp. 6338–6347, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/59dfa2df42d9e3d41f5b02bfc32229dd-Abstract.html>.
- Yann Ollivier. Ricci curvature of Markov chains on metric spaces. *Journal of Functional Analysis*, 256(3):810–864, 2009.
- Avik Pal, Max van Spengler, Guido Maria D’Amely di Melendugno, Alessandro Flaborea, Fabio Galasso, and Pascal Mettes. Compositional entailment learning for hyperbolic vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=3i13Gev2hV>.
- Varian Pappas, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. doi: 10.1073/pnas.2015509117. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2015509117>.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. *arXiv preprint arXiv:2406.01506*, 2024.
- Phillip Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The intrinsic dimension of images and its impact on learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.
- Rik Sarkar. Low distortion delaunay embedding of trees in hyperbolic plane. In Marc J. van Kreveld and Bettina Speckmann (eds.), *Graph Drawing - 19th International Symposium, GD 2011, Eindhoven, The Netherlands, September 21-23, 2011, Revised Selected Papers*, Lecture Notes in Computer Science, pp. 355–366. Springer, 2011. doi: 10.1007/978-3-642-25878-7\_34. URL [https://doi.org/10.1007/978-3-642-25878-7\\_34](https://doi.org/10.1007/978-3-642-25878-7_34).
- Aditya Sinha, Siqi Zeng, Makoto Yamada, and Han Zhao. Learning structured representations with hyperbolic embeddings. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 91220–91259. Curran Associates, Inc., 2024. doi: 10.52202/079017-2895. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/file/a5d2da376bab7624b3caeb9f78fcaa2f-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/a5d2da376bab7624b3caeb9f78fcaa2f-Paper-Conference.pdf).
- Alexandru Tifrea, Gary Bécigneul, and Octavian-Eugen Ganea. Poincaré glove: Hyperbolic word embeddings. In *International Conference on Learning Representations (ICLR)*, 2019.
- Menglin Yang, Ram Samarth B B, Aosong Feng, Bo Xiong, Jiahong Liu, Irwin King, and Rex Ying. Hyperbolic fine-tuning for large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=TkEdQv0bXB>.

## A FULL NUMERICAL TABLES

Every table is generated by script directly from the released per-cell result files; no value is hand-transcribed.

### A.1 WHAT EACH NULL PRESERVES

| null                                | preserves                               | destroys                                       |
|-------------------------------------|-----------------------------------------|------------------------------------------------|
| Gaussian, spectrum-matched          | $n, d$ , mean, covariance spectrum      | all higher-order structure                     |
| PC-permutation                      | each principal component's marginal     | dependence across components                   |
| Cluster-preserving (tested, unused) | each cluster's geometry, hub assignment | hub arrangement (invalid depth test: App. A.5) |
| Label shuffle (ablation)            | the point cloud itself                  | embedding-label correspondence                 |

Table 2: The nulls of the instrument and what each one preserves and destroys; “genuine” in the text always means beyond the first of these.

### A.2 ESTIMATOR CALIBRATION

| space                                                   | absolute $\delta$   | $\delta_{\text{norm}}$              |
|---------------------------------------------------------|---------------------|-------------------------------------|
| balanced binary tree (depth 10)                         | 0.000               | 0.000                               |
| $\mathbb{H}^2$ ( $K = -1$ ), region radius $R=2/4/8/16$ | 0.65/0.69/0.69/0.69 | 0.162/0.087/0.043/0.022             |
| uniform $S^{99}$ (chord / geodesic)                     | 0.25/0.37           | 0.143/0.179                         |
| iid Gaussian ( $n=1000$ ), $d=192, \dots, 1536$         | —                   | 0.104/0.081/0.074/0.061/0.055/0.046 |

Table 3: **A low raw  $\delta$  is not evidence of hierarchy.** Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of  $\mathbb{H}^2$ , spheres stay high, and iid Gaussians drift low as  $d$  grows, the dimension confound. Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze.

### A.3 MODEL PANEL

Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num\_classes=0, i.e. CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 13); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

| type                    | model          | params |
|-------------------------|----------------|--------|
| Supervised ViT (vision) | ViT-T (i21k)   | 5M     |
| Supervised ViT (vision) | ViT-S (i21k)   | 22M    |
| Supervised ViT (vision) | ViT-B (i21k)   | 86M    |
| Supervised ViT (vision) | ViT-L (i21k)   | 307M   |
| SSL (vision)            | DINO-B         | 86M    |
| SSL (vision)            | DINOv2-S       | 22M    |
| SSL (vision)            | DINOv2-B       | 86M    |
| SSL (vision)            | DINOv2-L       | 307M   |
| SSL (vision)            | DINOv2-G       | 1.1B   |
| Contrastive (vision)    | CLIP-B         | 86M    |
| Contrastive (vision)    | CLIP-L         | 307M   |
| Contrastive (vision)    | SigLIP-B       | 86M    |
| Causal LM               | GPT-2 S        | 117M   |
| Causal LM               | GPT-2 M        | 345M   |
| Causal LM               | GPT-2 L        | 774M   |
| Causal LM               | GPT-2 XL       | 1.5B   |
| Causal LM               | Pythia-410M    | 410M   |
| Causal LM               | Pythia-1B      | 1.0B   |
| Causal LM               | Pythia-2.8B    | 2.8B   |
| Causal LM               | OLMo-1B        | 1.0B   |
| Causal LM               | OLMo-7B        | 7.0B   |
| Text embedder           | BGE-base       | 110M   |
| Text embedder           | BGE-large      | 335M   |
| Text embedder           | GTE-base       | 110M   |
| Text embedder           | GTE-large      | 335M   |
| Text embedder           | GTE-Qwen2-1.5B | 1.5B   |
| Text embedder           | E5-base        | 110M   |
| Text embedder           | E5-large       | 335M   |

Table 4: The full model panel: 12 vision backbones (all in the census; the 10 with cached per-dataset features, i.e. all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, and 7 text embedders.

#### A.4 VISION CENSUS: EXCESS AND SIGNIFICANCE PER CELL

| model    | sup. IN ( $r$ ) | exc. DTD | exc. FMNIST | exc. MNIST | best-R (pp) | metric |
|----------|-----------------|----------|-------------|------------|-------------|--------|
| ViT-T    | −0.020 (199)    | −0.014   | +0.018°     | −0.026°    | +0.58       | either |
| ViT-S    | −0.020 (200)    | −0.019   | −0.033      | −0.037°    | +0.49       | either |
| ViT-B    | −0.022 (200)    | −0.021   | −0.042      | −0.013°    | +0.51       | either |
| ViT-L    | −0.027 (200)    | −0.031   | −0.051      | −0.040°    | +0.87       | either |
| DINO-B   | +0.000 (96)     | −0.027   | −0.007°     | −0.034°    | —           | —      |
| DINOv2-S | +0.017 (0)      | −0.018   | −0.031°     | −0.004°    | +0.68       | H      |
| DINOv2-B | +0.018 (0)      | −0.035   | −0.062      | −0.017°    | +1.17       | cos    |
| DINOv2-L | +0.006 (0)      | −0.042   | −0.035      | −0.014°    | +1.84       | cos    |
| DINOv2-G | +0.020 (0)      | −0.039   | −0.046      | −0.022°    | +2.54       | cos    |
| CLIP-B   | −0.023 (199)    | −0.024   | −0.035°     | −0.001°    | +0.67       | H      |
| CLIP-L   | −0.016 (198)    | −0.027   | −0.037°     | −0.008°    | +0.65       | H      |
| SigLIP-B | −0.015 (194)    | −0.023   | −0.042      | −0.044     | —           | —      |

Table 5: Census columns moved out of Table 1. sup. IN: ImageNet excess of the supremum statistic under the same Haar null (200 replicates;  $r$  = replicates above the real value; Table 31); exc. DTD/FMNIST/MNIST: per-dataset record excess on the remaining sets; best-R: few-shot advantage of the best zero-cost metric over Euclidean, mean over the four hierarchical datasets; metric: which of Poincaré (H) or cosine collects it (either: within 0.15pp; —: outside the 10-model task grid).

| model                    | $d$  | $\hat{\delta}$ | excess ( $z$ ) | reps above |
|--------------------------|------|----------------|----------------|------------|
| Barlow Twins (ResNet-50) | 2048 | 0.075          | −0.039 (−4.8)  | 20/20      |
| BYOL (ResNet-50)         | 2048 | 0.073          | −0.020 (−2.4)  | 20/20      |

Table 6: Backbones outside the ViT census, ImageNet centroid store: two ResNet-50 self-supervised models, under the spectrum null (20 replicates, percentile rank).

## A.5 WHAT THE EXCESS CERTIFIES: SYNTHETIC CALIBRATION

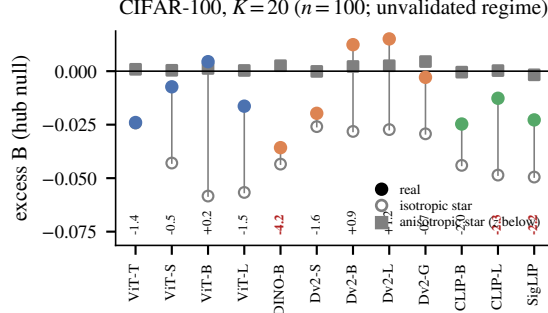


Figure 6: **The depth test on CIFAR-100 ( $K=20$ , five points per cluster): the unvalidated regime.** Excess B under the hub-randomizing null for the real centroids (filled) and their matched stars (hollow circle: isotropic, withdrawn; square: anisotropic); the number under each backbone is the  $z$  of real minus anisotropic star. In this regime pure synthetic stars are declared more hierarchical in 17% of runs (Table 9), so these readings are reported without certification.

**Real-data readings by regime.** ImageNet ( $n=1000$ , WordNet  $K=30$ ) lies in the validated regime: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ( $z \leq -2$ ) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less (Table 8, Figure 4a). CIFAR-100 ( $n=100$ ,  $K=20$ ) lies in the unvalidated regime, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with  $z \leq -2$ , none with  $z \geq +2$ ) are reported without certification (Figure 6). Validity domain of the test as run:  $n \approx 1000$ ; at  $n=100$  the false-alarm rate is 0% with  $K=6$  (17 points per cluster) and 17% with  $K \geq 12$ , so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the validated regime was adopted after it.

| synthetic cloud ( $n=1000$ , $d=768$ ) | $\hat{\delta}$ | spectrum null A |        | hub-randomizing null B |        |
|----------------------------------------|----------------|-----------------|--------|------------------------|--------|
|                                        |                | Gaussian        | Haar   | Gaussian               | Haar   |
| 30-cluster star, tight (0.1)           | 0.055          | -0.115          | -0.115 | -0.126                 | -0.058 |
| 30-cluster star, mid (0.3)             | 0.059          | -0.104          | -0.103 | -0.114                 | -0.049 |
| 30-cluster star, loose (0.6)           | 0.061          | -0.084          | -0.082 | -0.090                 | -0.033 |
| 2-level hierarchy $6 \times 5$ (0.3)   | 0.044          | -0.143          | -0.143 | -0.113                 | -0.112 |

Table 7: Calibration of the nulls on synthetic clouds (excess, means over seeds; within/between-cluster noise ratio in parentheses). A pure star (Gaussian clusters around Gaussian centers, no hierarchy) sits far below the spectrum null A under either construction (Gaussian coefficients, the paper’s; Haar-rotated coefficients, exact sample spectrum): null A certifies clustering, not depth. The hub-randomizing null B, which keeps every cluster intact and resamples only the hub configuration, also reports the star as “hierarchical”: a near-regular simplex of hubs already minimizes  $\hat{\delta}$ , so any hub randomization raises it. Null B is therefore not a valid depth test as constructed and is not used in the paper; a star-calibrated version (excess B relative to a matched star with the same cluster count and tightness) may still serve as a depth test, since under the Haar construction the two-level hierarchy sits 2–3 $\times$  below the star. That version is run on the real backbones in Table 8. Under the plain Gaussian null the mid-radius star scores -0.104 and the  $6 \times 5$  hierarchy -0.14: a star and a two-level tree differ by a fraction of either’s excess, which is why depth cannot be read from the spectrum excess alone.

| model    | CIFAR-100 ( $K=20$ ) |                  | ImageNet ( $K=30$ ) |                  |
|----------|----------------------|------------------|---------------------|------------------|
|          | isotropic star       | anisotropic star | isotropic star      | anisotropic star |
| ViT-T    | +0.000 (+0.0)        | -0.025 (-1.4)    | -0.015 (-1.5)       | -0.018 (-1.5)    |
| ViT-S    | +0.036 (+2.3)        | -0.008 (-0.5)    | -0.021 (-1.9)       | -0.028 (-2.4)    |
| ViT-B    | +0.063 (+4.5)        | +0.003 (+0.2)    | -0.026 (-2.9)       | -0.033 (-3.6)    |
| ViT-L    | +0.040 (+4.2)        | -0.017 (-1.5)    | -0.024 (-3.2)       | -0.033 (-4.0)    |
| DINO-B   | +0.008 (+0.9)        | -0.038 (-4.2)    | -0.002 (-0.2)       | -0.009 (-0.9)    |
| DINOv2-S | +0.006 (+0.6)        | -0.020 (-1.6)    | +0.011 (+0.9)       | +0.008 (+0.7)    |
| DINOv2-B | +0.041 (+5.4)        | +0.010 (+0.9)    | -0.011 (-0.9)       | -0.016 (-1.2)    |
| DINOv2-L | +0.042 (+6.8)        | +0.013 (+1.2)    | -0.024 (-2.3)       | -0.028 (-2.4)    |
| DINOv2-G | +0.026 (+3.8)        | -0.007 (-0.7)    | -0.024 (-1.6)       | -0.029 (-1.8)    |
| CLIP-B   | +0.019 (+1.8)        | -0.024 (-2.0)    | -0.006 (-0.7)       | -0.016 (-2.0)    |
| CLIP-L   | +0.036 (+5.7)        | -0.013 (-2.3)    | +0.004 (+0.5)       | -0.008 (-0.8)    |
| SigLIP-B | +0.027 (+2.8)        | -0.021 (-2.2)    | +0.001 (+0.0)       | -0.011 (-0.9)    |

**Table 8: The matched-star depth test on the real backbones: the star’s shape decides the verdict.** The test is validated at  $n=1000$  only (Table 9); CIFAR-100 readings ( $n=100$ ) are reported without certification. Certified on ImageNet ( $n=1000$ , the validated regime): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with  $z$  from  $-2.4$  to  $-4.0$ ; the withdrawn isotropic star reached  $z = +6.8$  on CIFAR-100. Depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame than the star), with  $z$  against the combined spread; 10 star seeds; hub-randomizing Haar null. Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction). Anisotropic star: within each superclass a Haar sample with the cloud’s own covariance. Frames: CIFAR-100 coarse labels; ImageNet WordNet cut. K sweep: CIFAR-100  $K=5$ : iso 0/12 at  $z \leq -2$ , 7/12 at  $z \geq +2$ ; aniso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=10$ : iso 0/12 at  $z \leq -2$ , 9/12 at  $z \geq +2$ ; aniso 1/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=20$ : iso 0/12 at  $z \leq -2$ , 8/12 at  $z \geq +2$ ; aniso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=10$ : iso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 2/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=30$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=60$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ .

| frame                           | $n$  | $K$ | power @ ratio 0.1 | 0.3  | 0.6  | star $z \leq -2$ | star $z \geq +2$ |
|---------------------------------|------|-----|-------------------|------|------|------------------|------------------|
| top-level, isotropic star       | 100  | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.13             |
| top-level, isotropic star       | 100  | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| leaf clusters, anisotropic star | 100  | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 100  | 12  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 100  | 20  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 12  | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 20  | 1.00              | 1.00 | 0.90 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 30  | 1.00              | 1.00 | 0.70 | 0.00             | 0.00             |

Table 9: **Power of the matched-star depth test on synthetic hierarchies.** Two- and three-level hierarchies (pooled) and pure stars,  $d=768$ , within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds. Power = fraction of hierarchy runs with  $z \leq -2$ ; star columns = false-alarm rates in each direction. Pre-set bar: power  $\geq 0.8$  for two- and three-level hierarchies at noise ratios  $\leq 0.3$  for both  $n$ , and false alarms  $\leq 5\%$  in each direction. Leaf frame: met at  $n=1000$  (power 1.00 at ratios  $\leq 0.3$  and 0.90 at 0.6; false alarms 0% and 0%), failed at  $n=100$ , where  $K \geq 12$  (under ten points per cluster) gives 17% false alarms at  $z \leq -2$ . Top-level frame: power 0.00, false alarms 16% pooled. Top-level frame: the test receives the  $K$  super-cluster labels, as the earlier isotropic test (withdrawn) did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the  $K$  hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star of Table 8 (configurations with more leaves than points are infeasible and omitted).

|            |         | CIFAR-100 ( $K=20, n=100$ : depth unvalidated) |                                  |           | ImageNet ( $K=30, n=1000$ : depth validated) |                                  |           |
|------------|---------|------------------------------------------------|----------------------------------|-----------|----------------------------------------------|----------------------------------|-----------|
| model      | input   | exc. ( $r$ )                                   | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ | exc. ( $r$ )                                 | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ |
| MERU ViT-S | images  | -0.020 (199)                                   | 0.117/0.117                      | -1.5      | -0.008° (186)                                | 0.088/0.088                      | -1.5      |
| CLIP ViT-S | images  | -0.028 (200)                                   | 0.106/0.107                      | -3.9      | -0.006° (171)                                | 0.110/0.088                      | -1.8      |
| MERU ViT-B | images  | -0.020 (200)                                   | 0.114/0.114                      | -2.2      | -0.010 (196)                                 | 0.087/0.087                      | -0.9      |
| CLIP ViT-B | images  | -0.030 (200)                                   | 0.105/0.103                      | -3.3      | -0.009° (187)                                | 0.107/0.085                      | -1.0      |
| MERU ViT-L | images  | -0.012° (195)                                  | 0.118/0.118                      | -0.3      | -0.007° (184)                                | 0.088/0.088                      | -0.5      |
| CLIP ViT-L | images  | -0.016 (199)                                   | 0.123/0.112                      | -1.7      | -0.010 (197)                                 | 0.104/0.084                      | -1.2      |
| MERU ViT-S | prompts | -0.022 (200)                                   | 0.086/0.086                      | -0.7      | -0.003° (155)                                | 0.085/0.086                      | +0.0      |
| CLIP ViT-S | prompts | -0.023 (200)                                   | 0.085/0.082                      | -0.8      | +0.006° (15)                                 | 0.093/0.087                      | +0.8      |
| MERU ViT-B | prompts | -0.012 (200)                                   | 0.096/0.096                      | -0.7      | -0.001° (132)                                | 0.089/0.090                      | +0.8      |
| CLIP ViT-B | prompts | -0.017 (200)                                   | 0.094/0.088                      | +0.3      | +0.008° (2)                                  | 0.100/0.092                      | +0.2      |
| MERU ViT-L | prompts | -0.006° (190)                                  | 0.102/0.102                      | +0.6      | +0.001° (75)                                 | 0.093/0.093                      | +0.5      |
| CLIP ViT-L | prompts | -0.012 (200)                                   | 0.098/0.092                      | +0.5      | +0.006° (5)                                  | 0.101/0.092                      | +0.4      |

Table 10: **Imposing the geometry does not create depth: MERU and its Euclidean twin under the census of record.** MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released Euclidean twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. ( $r$ ): excess of  $\hat{\delta}_{99.9}$  over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under Benjamini–Hochberg over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth  $z$ : the anisotropic matched-star test of Table 8 (10 star seeds; negative = more hierarchical above the frame than a star), validated at  $n=1000$  only (Table 9). On ImageNet images, the validated regime, the excess runs from  $-0.007$  to  $-0.010$  for MERU and from  $-0.006$  to  $-0.010$  for its twin across ViT-S/B/L; MERU’s Lorentz-metric  $\hat{\delta}_{99.9}$  differs from its Euclidean value by at most 0.0002; depth  $z$  runs from  $-1.5$  to  $-0.5$  for MERU and from  $-1.8$  to  $-1.0$  for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy.

## A.6 TRAINING INTERVENTIONS

**Clustered structure tracks training.** Two interventions move it (Figure 7; raw  $\delta$ , within architecture): fine-tuning on a task with no class hierarchy raises  $\delta$  by 19–91%, and  $\delta$  falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw  $\delta$  but near-Gaussian features, so they are not counted as evidence; excess-based versions of these interventions await stored point clouds. On matched architecture (ViT-B supervised/DINO/CLIP) all three objectives yield genuine form on most datasets, and the family separations emerge with scale rather than at fixed size.

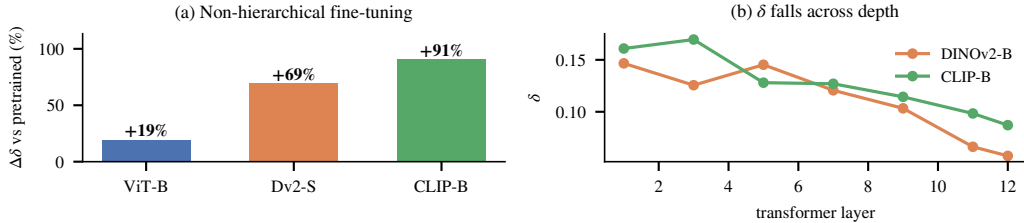


Figure 7: **The raw reading is learned: training interventions move it.** Raw  $\delta$ , within architecture. (a) Non-hierarchical fine-tuning raises  $\delta$  19–91%. (b)  $\delta$  falls across transformer depth (layers  $\geq 1$ ).

## A.7 TEXT CENSUS

**Extraction, anisotropy and templates.** Extraction is part of the measurement: left-padded batching moves GPT-2’s  $\hat{\delta}$  by up to 0.02, of the order of the fragile excesses; the padding-free protocol is

the record (Table 13). Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine. The only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record (Table 12). Causal-LM values are template-sensitive but the scale ordering is template-robust (Tables 36 and 37).

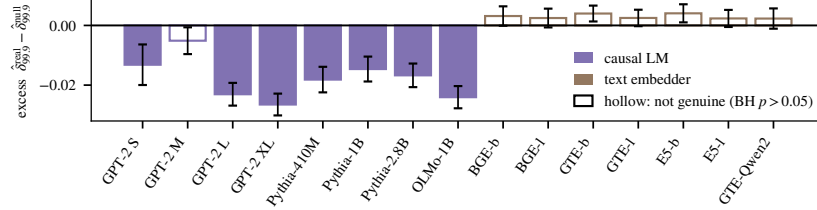


Figure 8: **In text, clustered structure depends on recipe and scale: robust at the larger GPT-2 sizes, fragile at the smaller.** Text census of record (padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates; error bars combine null and estimator s.d.; hollow bars are not genuine, BH  $p > 0.05$ ; OLMo-7B, not re-extracted, omitted). Pythia and OLMo-1B are genuine throughout; classic sentence embedders are not on this probe despite raw values equal to genuinely structured vision models.

| model          | type      | $d$  | $\delta_{\text{norm}}$ | null              | excess | $z$  |
|----------------|-----------|------|------------------------|-------------------|--------|------|
| GPT-2 S        | causal LM | 768  | $0.131 \pm 0.007$      | $0.083 \pm 0.002$ | +0.048 | +6.5 |
| GPT-2 M        | causal LM | 1024 | $0.091 \pm 0.004$      | $0.103 \pm 0.002$ | -0.013 | -2.7 |
| GPT-2 L        | causal LM | 1280 | $0.095 \pm 0.004$      | $0.129 \pm 0.006$ | -0.034 | -5.0 |
| GPT-2 XL       | causal LM | 1600 | $0.083 \pm 0.005$      | $0.131 \pm 0.007$ | -0.048 | -5.4 |
| Pythia-410M    | causal LM | 1024 | $0.098 \pm 0.005$      | $0.136 \pm 0.005$ | -0.037 | -5.4 |
| Pythia-1B      | causal LM | 2048 | $0.096 \pm 0.004$      | $0.131 \pm 0.004$ | -0.035 | -6.2 |
| Pythia-2.8B    | causal LM | 2560 | $0.098 \pm 0.004$      | $0.139 \pm 0.008$ | -0.041 | -4.6 |
| OLMo-1B        | causal LM | 2048 | $0.100 \pm 0.005$      | $0.101 \pm 0.002$ | -0.001 | -0.2 |
| OLMo-7B        | causal LM | 4096 | $0.093 \pm 0.005$      | $0.108 \pm 0.002$ | -0.015 | -2.8 |
| BGE-base       | embedder  | 768  | $0.121 \pm 0.006$      | $0.117 \pm 0.005$ | +0.005 | +0.6 |
| BGE-large      | embedder  | 1024 | $0.121 \pm 0.008$      | $0.120 \pm 0.003$ | +0.001 | +0.1 |
| GTE-base       | embedder  | 768  | $0.124 \pm 0.011$      | $0.113 \pm 0.003$ | +0.011 | +0.9 |
| GTE-large      | embedder  | 1024 | $0.120 \pm 0.012$      | $0.116 \pm 0.003$ | +0.004 | +0.4 |
| E5-base        | embedder  | 768  | $0.115 \pm 0.007$      | $0.112 \pm 0.005$ | +0.003 | +0.3 |
| E5-large       | embedder  | 1024 | $0.114 \pm 0.010$      | $0.113 \pm 0.003$ | +0.001 | +0.1 |
| GTE-Qwen2-1.5B | embedder  | 1536 | $0.107 \pm 0.004$      | $0.126 \pm 0.004$ | -0.019 | -3.1 |

Table 11: Text models under the calibrated protocol (1000 ImageNet class prompts):  $\delta_{\text{norm}}$ , spectrum-matched null (3 reps) and excess.

| model          | padding-free extraction, record (Haar $\times$ p99.9, 200 replicates) |        |         |      |              | original extraction (Table 11) |        |
|----------------|-----------------------------------------------------------------------|--------|---------|------|--------------|--------------------------------|--------|
|                | $\hat{\delta}_{99.9}$                                                 | excess | $r/200$ | $p$  | 2 $\times$ 2 | $\hat{\delta}$                 | excess |
| GPT-2 S        | 0.077                                                                 | -0.013 | 197     | .020 | ●○○○         | 0.131                          | +0.048 |
| GPT-2 M        | 0.052                                                                 | -0.005 | 170     | .154 | ○○○○         | 0.091                          | -0.013 |
| GPT-2 L        | 0.053                                                                 | -0.023 | 200     | .005 | ●●●●         | 0.095                          | -0.034 |
| GPT-2 XL       | 0.048                                                                 | -0.027 | 200     | .005 | ●●●●         | 0.083                          | -0.048 |
| Pythia-410M    | 0.067                                                                 | -0.018 | 200     | .005 | ●●●●         | 0.098                          | -0.037 |
| Pythia-1B      | 0.066                                                                 | -0.015 | 199     | .010 | ●●●●         | 0.096                          | -0.035 |
| Pythia-2.8B    | 0.066                                                                 | -0.017 | 200     | .005 | ●●●●         | 0.098                          | -0.041 |
| OLMo-1B        | 0.051                                                                 | -0.024 | 200     | .005 | ●●●●         | 0.100                          | -0.001 |
| OLMo-7B        | —                                                                     | —      | —       | —    | —            | 0.093                          | -0.015 |
| BGE-base       | 0.078                                                                 | +0.003 | 26      | .871 | ○○○○         | 0.121                          | +0.005 |
| BGE-large      | 0.078                                                                 | +0.002 | 49      | .756 | ○○○○         | 0.121                          | +0.001 |
| GTE-base       | 0.076                                                                 | +0.004 | 11      | .945 | ○○○○         | 0.124                          | +0.011 |
| GTE-large      | 0.075                                                                 | +0.003 | 36      | .821 | ○○○○         | 0.120                          | +0.004 |
| GTE-Qwen2-1.5B | 0.078                                                                 | +0.002 | 58      | .711 | ○○○○         | 0.107                          | -0.019 |
| E5-base        | 0.074                                                                 | +0.004 | 15      | .925 | ○○○○         | 0.115                          | +0.003 |
| E5-large       | 0.071                                                                 | +0.002 | 44      | .781 | ○○○○         | 0.114                          | +0.001 |

Table 12: **The text census of record: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.**  $r$  = replicates above the real value,  $p$  the left-tail add-one  $p$ -value; genuine = BH-corrected  $p \leq 0.05$  over the 15 models. The 2 $\times$ 2 column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine). OLMo-7B not re-run (exceeds the local GPU). The original extraction (left-padded batches, supremum statistic) is kept for comparison and is not directly comparable. Sign agreement with the 3-replicate original: 13/15. Three verdicts change between extractions, each with a known cause: GPT-2 S (above null in the left-padded original; Table 13), OLMo-1B (the original used a different checkpoint revision), and GTE-Qwen2 (genuine in the original with 3 replicates, -0.019; here excess +0.002,  $r = 58$ ,  $p = .711$ ; not used as evidence).

| batch size     | GPT-2 M             |                     | OLMo-1B             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|
|                | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ |
| 1 (no padding) | 0.0816              | 0.0817              | 0.0817              | 0.0817              |
| 16             | 0.0996              | 0.0997              | 0.0817              | 0.0816              |
| 32             | 0.0905              | 0.0905              | 0.0817              | 0.0817              |

Table 13: Where extraction-to-extraction variability comes from. Same 1000 prompts, same model; only batch size and precision vary. Precision is irrelevant (fp16 equals fp32 to four decimals). For GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves  $\hat{\delta}$  by up to 0.018; OLMo handles left padding correctly and is invariant. Padding-free (batch-size-1) extraction is therefore the protocol of Table 12.

## A.8 CLASS-COUNT VS HIERARCHY DEPTH

**Hierarchy depth, not class count.** Because  $C$  sets the quadruple population, we sweep ImageNet subsets  $C \in \{10, \dots, 1000\}$  in two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). At fixed  $C$ , random subsets are more exploitable than coherent ones and carry a larger excess at small  $C$  (CLIP-L at every  $C$ , DINOv2-L up to  $C=100$ , DINOv2-G up to  $C=20$ ; beyond that the two compositions converge; Table 14). Random subsets still score -0.08 to -0.14 at  $C=10$ , so MNIST’s flat verdict is not a class-count artifact, and DINOv2-G stays below its null at every  $C$  (excess -0.008 at  $C=1000$ ,  $r=200$ ), where the supremum had placed it at the null.

| model    | $C$  | random subsets (span the hierarchy) |        |       |        | WordNet-coherent (siblings) |        |       |        |
|----------|------|-------------------------------------|--------|-------|--------|-----------------------------|--------|-------|--------|
|          |      | $\hat{\delta}_{99.9}$               | excess | below | NC adv | $\hat{\delta}_{99.9}$       | excess | below | NC adv |
| DINOv2-L | 10   | 0.030                               | -0.134 | 5/5   | +1.04  | 0.068                       | -0.098 | 5/5   | +0.08  |
| DINOv2-L | 20   | 0.034                               | -0.091 | 5/5   | +1.32  | 0.071                       | -0.065 | 5/5   | +0.12  |
| DINOv2-L | 50   | 0.032                               | -0.046 | 5/5   | +1.58  | 0.050                       | -0.042 | 5/5   | +0.53  |
| DINOv2-L | 100  | 0.031                               | -0.032 | 5/5   | +1.46  | 0.044                       | -0.031 | 5/5   | +0.77  |
| DINOv2-L | 200  | 0.031                               | -0.022 | 5/5   | +1.45  | 0.041                       | -0.023 | 5/5   | +0.59  |
| DINOv2-L | 500  | 0.030                               | -0.016 | 2/2   | +1.01  | 0.031                       | -0.018 | 2/2   | +0.82  |
| DINOv2-L | 1000 | 0.030                               | -0.014 | 1/1   | +0.57  | —                           | —      | —     | —      |
| DINOv2-G | 10   | 0.029                               | -0.136 | 5/5   | +0.80  | 0.072                       | -0.095 | 5/5   | -0.56  |
| DINOv2-G | 20   | 0.037                               | -0.089 | 5/5   | +1.30  | 0.070                       | -0.067 | 5/5   | -0.18  |
| DINOv2-G | 50   | 0.039                               | -0.039 | 5/5   | +1.51  | 0.053                       | -0.042 | 5/5   | +0.26  |
| DINOv2-G | 100  | 0.041                               | -0.022 | 5/5   | +1.30  | 0.049                       | -0.028 | 5/5   | +0.36  |
| DINOv2-G | 200  | 0.036                               | -0.015 | 5/5   | +1.28  | 0.045                       | -0.020 | 5/5   | +0.27  |
| DINOv2-G | 500  | 0.034                               | -0.009 | 2/2   | +0.49  | 0.034                       | -0.013 | 2/2   | +0.22  |
| DINOv2-G | 1000 | 0.033                               | -0.008 | 1/1   | -0.13  | —                           | —      | —     | —      |
| CLIP-L   | 10   | 0.071                               | -0.080 | 5/5   | -0.08  | 0.099                       | -0.066 | 4/5   | -0.36  |
| CLIP-L   | 20   | 0.078                               | -0.047 | 5/5   | +0.20  | 0.113                       | -0.036 | 4/5   | -0.24  |
| CLIP-L   | 50   | 0.080                               | -0.027 | 5/5   | +0.34  | 0.113                       | -0.014 | 4/5   | -0.46  |
| CLIP-L   | 100  | 0.079                               | -0.024 | 5/5   | +0.21  | 0.095                       | -0.016 | 3/5   | -0.31  |
| CLIP-L   | 200  | 0.078                               | -0.019 | 5/5   | +0.00  | 0.086                       | -0.010 | 2/5   | -0.35  |
| CLIP-L   | 500  | 0.077                               | -0.012 | 2/2   | -0.27  | 0.081                       | -0.011 | 1/2   | -0.42  |
| CLIP-L   | 1000 | 0.076                               | -0.009 | 1/1   | -0.53  | —                           | —      | —     | —      |

Table 14: Class-count vs hierarchy-depth deconfound on ImageNet under the census of record (Haar spectrum-matched null, 99.9th-percentile statistic, 200 replicates; means over subset seeds: 5 for  $C \leq 200$ , 2 for  $C=500$ , 1 for  $C=1000$ ). “below” = subset seeds whose real value lies below the null at uncorrected  $p \leq 0.05$ ; NC adv = prototype-classifier advantage of the Poincaré readout in pp, from the unchanged task pipeline.

## A.9 TREE-MAP METRIC AND LINKAGE CONTROLS

Threshold sensitivity of the selection criterion of §5: the selected configuration is stable for degeneracy bounds in  $[0.45, 0.50]$  on ImageNet and  $[0.40, 0.50]$  on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

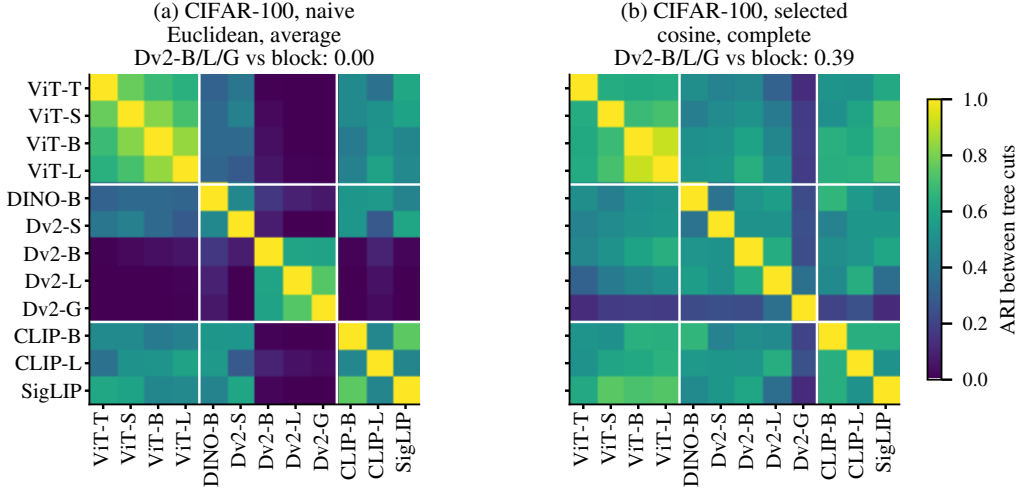


Figure 9: **The island is an artifact on CIFAR-100 too.** Same reading as Figure 5 on the 20 CIFAR-100 superclass cuts: naive (a) vs criterion-selected (b).

| dataset  | configuration              | max frac. | CPCC  | Dv2-B/L/G vs block | Dv2-S+DINO-B vs block | within block |
|----------|----------------------------|-----------|-------|--------------------|-----------------------|--------------|
| imagenet | euclid-average (deg.)      | 0.97      | 0.761 | 0.03               | 0.08                  | 0.64         |
| imagenet | euclid-complete (deg.)     | 0.83      | 0.606 | 0.03               | 0.17                  | 0.44         |
| imagenet | euclid-ward                | 0.48      | 0.464 | 0.15               | 0.38                  | 0.53         |
| imagenet | cosine-average (selected)  | 0.40      | 0.691 | 0.38               | 0.35                  | 0.48         |
| imagenet | cosine-complete            | 0.32      | 0.539 | 0.13               | 0.17                  | 0.50         |
| imagenet | cosine-ward                | 0.37      | 0.464 | 0.17               | 0.39                  | 0.55         |
| cifar100 | euclid-average (deg.)      | 0.80      | 0.798 | 0.00               | 0.41                  | 0.57         |
| cifar100 | euclid-complete (deg.)     | 0.66      | 0.722 | 0.17               | 0.40                  | 0.54         |
| cifar100 | euclid-ward                | 0.33      | 0.534 | 0.37               | 0.65                  | 0.63         |
| cifar100 | cosine-average (deg.)      | 0.53      | 0.777 | 0.30               | 0.54                  | 0.58         |
| cifar100 | cosine-complete (selected) | 0.39      | 0.698 | 0.39               | 0.51                  | 0.63         |
| cifar100 | cosine-ward                | 0.14      | 0.584 | 0.56               | 0.62                  | 0.61         |

Table 15: Tree-map configurations with the two quantities that drive the selection criterion of §5: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix. Configurations with max-cluster fraction  $> 0.5$  are excluded; among the rest, highest CPCC selects the marked row. Mean pairwise ARIs at the reference cut shown for all configurations. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes: the island is made by chaining.

| dataset   | configuration   | ARI at the cut | cophenetic corr. | triplet agreement |
|-----------|-----------------|----------------|------------------|-------------------|
| ImageNet  | euclid-average  | 0.03 / 0.64    | 0.36 / 0.80      | 0.47 / 0.74       |
| ImageNet  | euclid-complete | 0.03 / 0.44    | 0.20 / 0.66      | 0.41 / 0.65       |
| ImageNet  | euclid-ward     | 0.16 / 0.53    | 0.62 / 0.84      | 0.67 / 0.85       |
| ImageNet  | cosine-average  | 0.39 / 0.48    | 0.48 / 0.78      | 0.77 / 0.76       |
| ImageNet  | cosine-complete | 0.13 / 0.49    | 0.18 / 0.72      | 0.36 / 0.71       |
| ImageNet  | cosine-ward     | 0.17 / 0.55    | 0.59 / 0.85      | 0.69 / 0.86       |
| CIFAR-100 | euclid-average  | 0.00 / 0.57    | 0.15 / 0.78      | 0.41 / 0.74       |
| CIFAR-100 | euclid-complete | 0.17 / 0.54    | 0.23 / 0.72      | 0.41 / 0.66       |
| CIFAR-100 | euclid-ward     | 0.37 / 0.63    | 0.63 / 0.77      | 0.70 / 0.79       |
| CIFAR-100 | cosine-average  | 0.30 / 0.58    | 0.42 / 0.72      | 0.51 / 0.64       |
| CIFAR-100 | cosine-complete | 0.39 / 0.63    | 0.46 / 0.64      | 0.50 / 0.61       |
| CIFAR-100 | cosine-ward     | 0.56 / 0.61    | 0.74 / 0.73      | 0.76 / 0.76       |

Table 16: **The island is a property of the cut and of the Euclidean configurations.** Each entry: DINOv2-B/L/G vs the supervised+contrastive block / within the block. ARI between cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters; Pearson correlation between the two trees’ cophenetic distance vectors; agreement on which pair merges first over  $10^4$  random class triplets. Dendrograms rebuilt from the census cache with the six configurations of Table 15.

| model    | euclid-avg (naive, degen.) | euclid-ward | cosine-avg (degen. C100) | cosine-complete ← sel. | cosine-ward |
|----------|----------------------------|-------------|--------------------------|------------------------|-------------|
| ViT-T    | 0.31                       | 0.38        | 0.29                     | 0.33                   | 0.33        |
| ViT-S    | 0.35                       | 0.56        | 0.37                     | 0.47                   | 0.56        |
| ViT-B    | 0.39                       | 0.61        | 0.43                     | 0.45                   | 0.57        |
| ViT-L    | 0.38                       | 0.55        | 0.49                     | 0.48                   | 0.60        |
| DINO-B   | 0.22                       | 0.48        | 0.28                     | 0.38                   | 0.54        |
| DINOv2-S | 0.12                       | 0.51        | 0.32                     | 0.33                   | 0.49        |
| DINOv2-B | 0.02                       | 0.47        | 0.35                     | 0.42                   | 0.51        |
| DINOv2-L | 0.01                       | 0.32        | 0.25                     | 0.50                   | 0.48        |
| DINOv2-G | 0.00                       | 0.19        | 0.08                     | 0.20                   | 0.52        |
| CLIP-B   | 0.18                       | 0.48        | 0.18                     | 0.37                   | 0.48        |
| CLIP-L   | 0.35                       | 0.52        | 0.32                     | 0.41                   | 0.54        |
| SigLIP-B | 0.22                       | 0.49        | 0.22                     | 0.37                   | 0.52        |

Table 17: Recovery of the true CIFAR-100 superclasses (ARI of the 20-cluster cut) per model and configuration, the quantity behind the corrected picture of §5. Degenerate configurations marked; the criterion-selected configuration is cosine-complete. Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52.

#### A.10 THE COROLLARY: DIAGNOSING EXPLOITABILITY

**One parameter-free projection and two tasks serve the corollary.** The parameter-free map  $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$  centers features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$ , and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

**$\delta_{\text{norm}}$  predicts the gain within datasets.** Within each dataset the Poincaré advantage tracks  $\delta_{\text{norm}}$  (NC  $r = -0.83$  on ImageNet) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD (Appendix A.13). Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test (Appendix A.13).

**The training objective picks the metric.** The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2,

Poincaré  $\approx$  cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to +1.3pp even after L2 normalization (Figure 10 in Appendix A.13; held-out selection and policy comparison in this appendix). The rule is read off the families it is scored on: it describes them rather than validating a policy.

**A diagnostic, not a policy.** Khrulkov et al. (2020) estimate  $c \approx 0.33$  from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on  $\delta_{\text{norm}}$  is the weakest policy tested (this appendix), and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Appendix A.18). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

#### A.11 TASK ACCURACIES (REGENERATED)

| model    | IN        | C100      | C10       | DTD       | FMNIST    | MNIST     |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ViT-T    | 68.1/67.2 | 29.3/28.2 | 87.5/87.4 | 63.7/62.5 | 78.5/78.4 | 75.9/74.0 |
| ViT-S    | 77.7/77.5 | 74.9/74.6 | 93.4/93.2 | 69.3/68.5 | 80.7/80.4 | 84.8/84.1 |
| ViT-B    | 81.6/81.7 | 77.3/77.0 | 94.5/94.0 | 70.8/70.8 | 81.3/81.6 | 81.6/80.5 |
| ViT-L    | 86.1/86.2 | 85.3/85.2 | 97.5/97.5 | 71.8/71.5 | 83.7/83.5 | 84.2/83.6 |
| DINOv2-S | 74.7/74.4 | 73.4/74.4 | 95.0/95.2 | 73.4/73.4 | 82.2/82.2 | 83.4/81.9 |
| DINOv2-B | 78.9/78.9 | 87.4/87.9 | 97.1/97.4 | 74.2/74.7 | 83.2/82.8 | 80.6/78.8 |
| DINOv2-L | 79.8/80.4 | 85.6/87.7 | 97.9/98.3 | 74.4/75.2 | 84.4/84.5 | 74.1/72.7 |
| DINOv2-G | 78.4/78.3 | 89.8/91.0 | 97.8/98.2 | 72.9/73.1 | 85.9/85.8 | 79.3/78.6 |
| CLIP-B   | 61.4/59.9 | 65.6/65.1 | 87.9/88.3 | 63.7/63.2 | 75.9/75.1 | 70.1/66.6 |
| CLIP-L   | 75.0/74.5 | 74.6/74.5 | 95.9/95.9 | 68.6/69.0 | 76.2/75.9 | 85.2/86.1 |

Table 18: Nearest-centroid accuracy (%), Euclidean/Poincaré, regenerated from the audited reruns.

| model    | IN        | C100      | C10       | DTD       | FMNIST    | MNIST     |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|
| ViT-T    | 97.5/97.6 | 62.4/63.1 | 87.6/88.1 | 83.3/83.9 | 81.6/81.9 | 74.3/74.6 |
| ViT-S    | 98.6/98.7 | 92.2/92.4 | 92.0/92.5 | 85.4/86.0 | 83.7/83.5 | 80.5/81.4 |
| ViT-B    | 98.9/99.0 | 92.5/92.7 | 92.7/93.0 | 85.4/86.2 | 83.4/83.3 | 76.8/77.6 |
| ViT-L    | 99.4/99.4 | 94.8/95.5 | 95.0/95.8 | 85.6/86.5 | 84.4/84.7 | 80.8/81.6 |
| DINOv2-S | 97.4/97.9 | 94.3/95.1 | 92.9/93.5 | 89.8/90.5 | 85.4/85.5 | 80.2/80.8 |
| DINOv2-B | 97.0/97.8 | 96.5/97.2 | 94.7/95.4 | 89.2/90.3 | 85.9/85.9 | 76.6/77.1 |
| DINOv2-L | 96.2/97.1 | 96.1/97.1 | 94.1/95.1 | 87.1/88.4 | 87.8/87.9 | 70.5/71.1 |
| DINOv2-G | 96.2/97.1 | 95.9/97.2 | 92.3/93.8 | 87.2/88.5 | 89.0/89.2 | 72.9/73.8 |
| CLIP-B   | 96.8/96.8 | 87.7/88.9 | 89.5/90.1 | 82.2/83.1 | 78.8/78.9 | 70.9/69.7 |
| CLIP-L   | 98.4/98.6 | 90.0/90.8 | 92.4/93.0 | 85.2/86.2 | 76.8/77.3 | 83.7/84.7 |

Table 19: 5-way 5-shot accuracy (%), Euclidean/Poincaré, 1000 paired episodes.

## A.12 METRIC CONTROLS

| model    | IN    | C100  | C10   | DTD   | FMNIST | MNIST |
|----------|-------|-------|-------|-------|--------|-------|
| ViT-T    | -0.01 | +1.01 | +0.66 | -0.23 | +0.21  | +0.64 |
| ViT-S    | +0.17 | +0.59 | +0.56 | -0.03 | +0.28  | +0.34 |
| ViT-B    | +0.20 | +0.56 | +0.49 | +0.28 | +0.03  | +0.19 |
| ViT-L    | +0.11 | +1.17 | +1.25 | +0.13 | +0.00  | +0.08 |
| DINOv2-S | +0.68 | +0.80 | -0.36 | +0.30 | +0.02  | -0.07 |
| DINOv2-B | +1.40 | +1.11 | +0.27 | +1.48 | +0.21  | +0.17 |
| DINOv2-L | +2.13 | +1.80 | +0.92 | +2.40 | +0.29  | +0.25 |
| DINOv2-G | +2.57 | +2.44 | +2.51 | +2.63 | +0.38  | +0.30 |
| CLIP-B   | -0.04 | -0.03 | -0.19 | -0.23 | +0.06  | +0.04 |
| CLIP-L   | +0.01 | +0.12 | +0.16 | -0.08 | +0.12  | +0.08 |

Table 20: Few-shot advantage of cosine over Euclidean (pp).

| model    | IN    | C100  | C10   | DTD   | FMNIST | MNIST |
|----------|-------|-------|-------|-------|--------|-------|
| ViT-T    | +0.01 | +0.45 | +0.19 | +0.32 | +0.10  | +0.52 |
| ViT-S    | -0.02 | +0.16 | +0.27 | +0.25 | +0.08  | +0.65 |
| ViT-B    | -0.04 | +0.12 | +0.18 | +0.19 | +0.09  | +0.51 |
| ViT-L    | -0.03 | +0.31 | +0.35 | +0.16 | +0.17  | +0.42 |
| DINOv2-S | +0.03 | +0.15 | +0.13 | +0.04 | +0.07  | +0.45 |
| DINOv2-B | +0.04 | +0.09 | +0.03 | +0.01 | +0.04  | +0.48 |
| DINOv2-L | +0.02 | +0.09 | -0.00 | +0.03 | +0.03  | +0.43 |
| DINOv2-G | +0.03 | +0.13 | -0.02 | +0.01 | +0.04  | +0.56 |
| CLIP-B   | -0.00 | +0.44 | +0.20 | +0.36 | +0.17  | +0.23 |
| CLIP-L   | +0.00 | +0.39 | +0.25 | +0.36 | +0.33  | +0.58 |

Table 21: Few-shot advantage of the radial-rescaling control over Euclidean (pp): the map without the metric changes nothing.

## A.13 STATISTICS

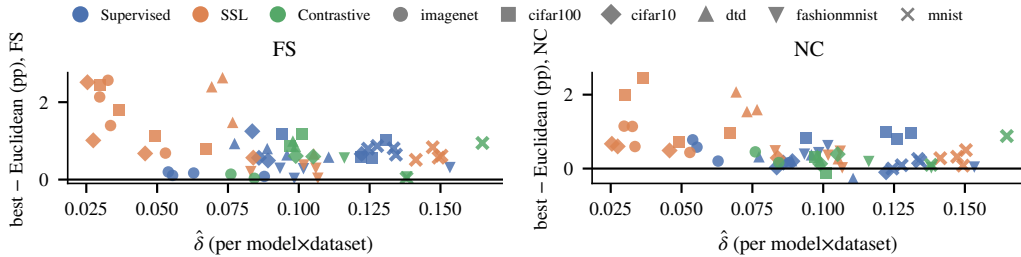


Figure 10: **More tree-like cells carry larger zero-cost metric gains.** Best zero-cost metric minus Euclidean per cell vs  $\delta_{\text{norm}}$  (FS left, NC right). Within-dataset correlations  $-0.60$  to  $-0.83$  (FS) and  $-0.61$  to  $-0.86$  (NC); pooled correlations are family-driven and not used as evidence (Appendix A.13).

| measure                          | raw   | null mean | $\tau_{0.05}$ | calibrated | null-centered | pairs $p < 0.05$ |
|----------------------------------|-------|-----------|---------------|------------|---------------|------------------|
| mutual-kNN ( $k=10$ ), Euclidean | 0.472 | 0.010     | 0.015         | 0.464      | 0.462         | 66/66            |
| mutual-kNN, Poincaré             | 0.425 | 0.010     | 0.019         | 0.414      | 0.415         | 66/66            |
| linear CKA, Euclidean            | 0.633 | 0.108     | 0.111         | 0.577      | 0.524         | 66/66            |
| linear CKA, Poincaré             | 0.640 | 0.108     | 0.111         | 0.586      | 0.532         | 66/66            |

Table 22: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with  $K=200$  permutations of the class correspondence and  $\alpha=0.05$ , scalar (no layer search):  $\tau_{0.05}$  is the  $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9),  $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$  (eq. 10), calibrated =  $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$  (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level  $k/(n-1) = 10/999$ .

| quantity  | dataset   | pooled $r$ | demeaned $r$ | ViT   | DINOv2 | CLIP |
|-----------|-----------|------------|--------------|-------|--------|------|
| NC H-R    | ImageNet  | -0.83      | -0.73        | -0.99 | -0.77  | (-1) |
|           | CIFAR-100 | -0.87      | -0.45        | -0.68 | -0.33  | (-1) |
|           | CIFAR-10  | -0.63      | -0.31        | +0.01 | -0.98  | (+1) |
|           | DTD       | -0.46      | -0.57        | -0.78 | -0.76  | (+1) |
| FS H-R    | ImageNet  | -0.57      | -0.73        | -0.04 | -0.97  | (-1) |
|           | CIFAR-100 | -0.67      | -0.63        | -0.41 | -0.96  | (+1) |
|           | CIFAR-10  | -0.78      | -0.64        | -0.22 | -0.79  | (-1) |
|           | DTD       | -0.71      | -0.86        | -0.95 | -0.98  | (+1) |
| FS best-R | ImageNet  | -0.64      | -0.71        | -0.57 | -0.90  | (-1) |
|           | CIFAR-100 | -0.83      | -0.82        | -0.58 | -0.96  | (+1) |
|           | CIFAR-10  | -0.63      | -0.55        | -0.26 | -0.65  | (-1) |
|           | DTD       | -0.60      | -0.70        | -0.95 | -0.88  | (+1) |

Table 23: Family-identity control for the  $\delta_{\text{norm}}$ -gain correlations (within dataset, 10 backbones = 3 families with nested scales). Demeaned: both variables centered within family before correlating; ViT/DINOv2: within-family correlations (4 models each); CLIP has only two models, so its within-family value (parenthesized sign) is not informative. Without DINOv2 (six backbones) the NC correlation holds on ImageNet and CIFAR-100 ( $-0.87/-0.65$ ) but not on CIFAR-10/DTD ( $+0.23/-0.34$ ). The within-dataset prediction survives the family control; pooled cross-dataset correlations of the best-metric advantage do not ( $-0.31 \rightarrow -0.18$  over all 60 cells,  $-0.33 \rightarrow -0.04$  on hierarchical cells) and are therefore family-driven.

| model    | IN            | C100          | C10           | DTD     | FMNIST       | MNIST         |
|----------|---------------|---------------|---------------|---------|--------------|---------------|
| ViT-T    | $10^{-22}(-)$ | $10^{-5}(-)$  | 0.19(-)       | 0.03(-) | 0.38(-)      | $10^{-13}(-)$ |
| ViT-S    | 0.09(-)       | 0.10(-)       | 0.02(-)       | 0.10(-) | 0.08(-)      | $10^{-4}(-)$  |
| ViT-B    | 0.21(+)       | 0.07(-)       | $10^{-5}(-)$  | 1.00(-) | 0.08(+)      | $10^{-7}(-)$  |
| ViT-L    | 0.58(+)       | 0.59(-)       | 0.76(+)       | 0.69(-) | 0.12(-)      | 0.00(-)       |
| DINOv2-S | 0.00(-)       | $10^{-6}(+)$  | 0.14(+)       | 1.00(+) | 1.00(-)      | $10^{-11}(-)$ |
| DINOv2-B | 0.57(+)       | $10^{-4}(+)$  | $10^{-4}(+)$  | 0.31(+) | 0.01(-)      | $10^{-16}(-)$ |
| DINOv2-L | $10^{-14}(+)$ | $10^{-38}(+)$ | $10^{-8}(+)$  | 0.11(+) | 0.40(+)      | $10^{-12}(-)$ |
| DINOv2-G | 0.10(-)       | $10^{-16}(+)$ | $10^{-13}(+)$ | 0.81(+) | 0.72(-)      | 0.00(-)       |
| CLIP-B   | $10^{-40}(-)$ | 0.07(-)       | $10^{-4}(+)$  | 0.46(-) | $10^{-4}(-)$ | $10^{-45}(-)$ |
| CLIP-L   | $10^{-8}(-)$  | 0.66(-)       | 0.59(-)       | 0.45(+) | 0.09(-)      | $10^{-7}(+)$  |

Table 24: McNemar  $p$ -values for NC test-set decisions, Poincaré vs Euclidean (order of magnitude when  $p < 10^{-3}$ ; sign of the NC advantage in parentheses, so significance is readable with its direction).

| task | dataset            | $r$   | CI95           |
|------|--------------------|-------|----------------|
| NC   | imagenet           | -0.83 | [-0.97, -0.65] |
| NC   | cifar100           | -0.87 | [-0.97, -0.78] |
| NC   | cifar10            | -0.63 | [-0.97, -0.06] |
| NC   | dtd                | -0.46 | [-0.90, +0.22] |
| FS   | imagenet           | -0.57 | [-0.91, +0.14] |
| FS   | cifar100           | -0.67 | [-0.95, -0.15] |
| FS   | cifar10            | -0.78 | [-0.93, -0.23] |
| FS   | dtd                | -0.71 | [-0.94, -0.32] |
| NC   | POOLED(cluster-bs) | -0.45 | [-0.60, -0.36] |
| FS   | POOLED(cluster-bs) | -0.18 | [-0.50, +0.12] |

Table 25: Bootstrap CI95 for the correlations between  $\delta_{\text{norm}}$  and the projection advantage (within-dataset: 10k resamples of the 10 models; pooled: cluster bootstrap over datasets).

| policy (test advantage over Euclidean, pp) | all 60 cells | hierarchical (40) |
|--------------------------------------------|--------------|-------------------|
| always cosine                              | +0.23        | +0.28             |
| objective rule (zero validation)           | +0.28        | +0.41             |
| $\delta$ -gated objective rule             | +0.08        | +0.13             |
| validation-picked per cell                 | +0.63        | +0.78             |
| test oracle (reference)                    | +0.64        | +0.78             |

Table 26: Metric-selection policies evaluated on held-out episodes (500 validation / 500 test per cell, paper protocol). Validation picking retains the oracle advantage; the zero-validation objective rule modestly beats always-cosine.

| NC policy (advantage over Euclidean, pp) | all 60 | flat (20) | hierarchical (40) |
|------------------------------------------|--------|-----------|-------------------|
| always cosine                            | +0.48  | +0.24     | +0.59             |
| always Poincaré                          | -0.24  | -0.70     | -0.00             |
| gated rule (flat→R)                      | +0.34  | +0.00     | +0.51             |
| $\delta$ -gated rule                     | +0.19  | +0.00     | +0.29             |

Table 27: NC metric-selection policies. The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere.

#### A.14 PROJECTION-TARGET SENSITIVITY

| target $t$        | 0.25  | 0.35  | 0.41  | 0.45  | 0.50  | 0.58  | 0.71  | 0.95  |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| FS adv. mean (pp) | +0.88 | +0.87 | +0.87 | +0.86 | +0.84 | +0.82 | +0.76 | +0.56 |
| NC adv. mean (pp) | -0.30 | -0.25 | -0.23 | -0.21 | -0.17 | -0.12 | -0.01 | +0.15 |
| % cells FS > 0    | 97    | 100   | 100   | 100   | 100   | 100   | 100   | 100   |

Table 28: Sensitivity to the projection target  $t$  (36 hierarchical cells). The headline  $t = 1/\sqrt{2} \approx 0.71$  was fixed a priori and is conservative for FS.

## A.15 ESTIMATOR ROBUSTNESS

| model                                                                         | data | excess vs quadruple budget ( $10^4 \dots 2 \times 10^5$ ) |         |                |        |                                  |        |        | p99.9 @ $5 \times 10^5$ |
|-------------------------------------------------------------------------------|------|-----------------------------------------------------------|---------|----------------|--------|----------------------------------|--------|--------|-------------------------|
| ViT-L                                                                         | IN   | -0.026                                                    | -0.024  | -0.027         | -0.030 | -0.034                           | -0.035 | -0.018 |                         |
| DINOv2-L                                                                      | IN   | -0.027                                                    | -0.019  | -0.024         | -0.011 | -0.004                           | -0.005 | -0.024 |                         |
| CLIP-B                                                                        | IN   | -0.008                                                    | -0.012  | -0.023         | -0.025 | -0.025                           | -0.031 | -0.007 |                         |
| ViT-L                                                                         | C100 | -0.045                                                    | -0.047  | -0.047         | -0.049 | -0.052                           | -0.053 | -0.033 |                         |
| DINOv2-L                                                                      | C100 | -0.071                                                    | -0.055  | -0.054         | -0.050 | -0.052                           | -0.055 | -0.069 |                         |
| CLIP-B                                                                        | C100 | -0.045                                                    | -0.047  | -0.050         | -0.049 | -0.052                           | -0.051 | -0.030 |                         |
| ViT-L                                                                         | DTD  | -0.064                                                    | -0.070  | -0.076         | -0.083 | -0.084                           | -0.084 | -0.054 |                         |
| DINOv2-L                                                                      | DTD  | -0.068                                                    | -0.072  | -0.075         | -0.083 | -0.084                           | -0.084 | -0.064 |                         |
| CLIP-B                                                                        | DTD  | -0.048                                                    | -0.055  | -0.059         | -0.065 | -0.066                           | -0.066 | -0.035 |                         |
| Centroid bootstrap (30 resamples of the per-class images): s.d. of the excess |      |                                                           |         |                |        |                                  |        |        |                         |
| ViT-L                                                                         | C100 | excess                                                    | -0.0493 | bootstrap s.d. | 0.0029 | (30 resamples, 500 images/class) |        |        |                         |
| DINOv2-L                                                                      | C100 | excess                                                    | -0.0493 | bootstrap s.d. | 0.0022 | (30 resamples, 500 images/class) |        |        |                         |
| DINOv2-G                                                                      | C100 | excess                                                    | -0.0778 | bootstrap s.d. | 0.0020 | (30 resamples, 500 images/class) |        |        |                         |
| CLIP-B                                                                        | C100 | excess                                                    | -0.0476 | bootstrap s.d. | 0.0017 | (30 resamples, 500 images/class) |        |        |                         |
| ViT-L                                                                         | DTD  | excess                                                    | -0.0803 | bootstrap s.d. | 0.0049 | (30 resamples, 80 images/class)  |        |        |                         |
| DINOv2-L                                                                      | DTD  | excess                                                    | -0.0878 | bootstrap s.d. | 0.0048 | (30 resamples, 80 images/class)  |        |        |                         |
| DINOv2-G                                                                      | DTD  | excess                                                    | -0.0863 | bootstrap s.d. | 0.0051 | (30 resamples, 80 images/class)  |        |        |                         |
| CLIP-B                                                                        | DTD  | excess                                                    | -0.0581 | bootstrap s.d. | 0.0054 | (30 resamples, 80 images/class)  |        |        |                         |

Table 29: Estimator robustness. Top: on CIFAR-100 and DTD the excess is stable once the quadruple budget reaches  $10^5$  (the paper uses  $5 \times 10^5$ ); on ImageNet ( $C=1000$ ) every sign holds at every budget but magnitudes drift by up to 0.02 (DINOv2-L toward its null, ViT-L and CLIP-B away from it), so ImageNet excess magnitudes carry a budget caveat. The 99.9th-percentile variant of the four-point statistic, robust to the supremum, is negative wherever the supremum excess is. Bottom: resampling the per-class images that form each centroid (reported on CIFAR-100 and DTD) moves the excess by an order of magnitude less than its size.

| model    | IN      |     |      | C100    |     |      | C10     |     |      | DTD    |     |      | FMNIST  |     |      | MNIST   |     |      |
|----------|---------|-----|------|---------|-----|------|---------|-----|------|--------|-----|------|---------|-----|------|---------|-----|------|
|          | exc.    | r   | p    | exc.    | r   | p    | exc.    | r   | p    | exc.   | r   | p    | exc.    | r   | p    | exc.    | r   | p    |
| ViT-T    | -0.005° | 169 | .159 | -0.006° | 151 | .249 | -0.021° | 172 | .144 | -0.014 | 197 | .020 | +0.018° | 34  | .831 | -0.026° | 168 | .164 |
| ViT-S    | -0.012  | 200 | .005 | -0.000° | 111 | .448 | -0.056  | 200 | .005 | -0.019 | 199 | .010 | -0.033  | 196 | .025 | -0.037° | 181 | .100 |
| ViT-B    | -0.016  | 200 | .005 | +0.006° | 43  | .786 | -0.050  | 200 | .005 | -0.021 | 200 | .005 | -0.042  | 200 | .005 | -0.013° | 142 | .294 |
| ViT-L    | -0.016  | 200 | .005 | -0.020  | 200 | .005 | -0.060  | 200 | .005 | -0.031 | 200 | .005 | -0.051  | 200 | .005 | -0.040° | 193 | .040 |
| DINO-B   | -0.020  | 200 | .005 | -0.032  | 200 | .005 | -0.061  | 200 | .005 | -0.027 | 200 | .005 | -0.007° | 127 | .368 | -0.034° | 188 | .065 |
| DINOv2-S | -0.016  | 200 | .005 | -0.031  | 200 | .005 | -0.071  | 200 | .005 | -0.018 | 200 | .005 | -0.031° | 193 | .040 | -0.004° | 112 | .443 |
| DINOv2-B | -0.017  | 200 | .005 | -0.031  | 200 | .005 | -0.112  | 200 | .005 | -0.035 | 200 | .005 | -0.062  | 200 | .005 | -0.017° | 147 | .269 |
| DINOv2-L | -0.014  | 200 | .005 | -0.035  | 200 | .005 | -0.128  | 200 | .005 | -0.042 | 200 | .005 | -0.035  | 197 | .020 | -0.014° | 132 | .343 |
| DINOv2-G | -0.008  | 200 | .005 | -0.040  | 200 | .005 | -0.130  | 200 | .005 | -0.039 | 200 | .005 | -0.046  | 199 | .010 | -0.022° | 158 | .214 |
| CLIP-B   | -0.007° | 172 | .144 | -0.019  | 200 | .005 | -0.036  | 195 | .030 | -0.024 | 200 | .005 | -0.035° | 181 | .100 | -0.001° | 104 | .483 |
| CLIP-L   | -0.009  | 197 | .020 | -0.018  | 200 | .005 | -0.043  | 199 | .010 | -0.027 | 200 | .005 | -0.037° | 188 | .065 | -0.008° | 126 | .373 |
| SigLIP-B | -0.008° | 189 | .060 | -0.022  | 200 | .005 | -0.050  | 199 | .010 | -0.023 | 200 | .005 | -0.042  | 198 | .015 | -0.044  | 198 | .015 |

Table 30: **The census of record per cell: Haar-rotated spectrum-matched null, 99.9th-percentile statistic, 200 replicates.** Excess over the null mean;  $r$  = replicates above the real value;  $p$  =  $(1 + \#\{\text{null} \leq \text{real}\})/201$ ; genuine = Benjamini–Hochberg-corrected  $p \leq 0.05$  over the 72 cells (°: not genuine). Genuine cells: 49/72 overall, 18/24 on ImageNet+CIFAR-100. Sign agreement with the original 5-replicate Gaussian  $\times$  supremum census on the 72 comparable cells: 67/72. Every cell on the census cache (one centroid source). The four null  $\times$  statistic verdicts per cell are in Table 31.

| model    | IN   | C100 | C10  | DTD  | FMNIST | MNIST |
|----------|------|------|------|------|--------|-------|
| ViT-T    | ○●○○ | ○●○○ | ○○○○ | ●●●● | ○○○○   | ○○○○  |
| ViT-S    | ●●●● | ○●○○ | ●●●● | ●●●● | ●●●●   | ○○○○  |
| ViT-B    | ●●●● | ○○○○ | ●●●● | ●●●● | ●●●●   | ○○○○  |
| ViT-L    | ●●●● | ●●●● | ●●●● | ●●●● | ●●●●   | ○○●●  |
| DINO-B   | ●○○○ | ●●●● | ●●●● | ●●●● | ○○○○   | ○○○○  |
| DINOv2-S | ●○○○ | ●●●● | ●●●● | ●●●● | ○○●●   | ○○○○  |
| DINOv2-B | ●○○○ | ○●○○ | ●●●● | ●●●● | ●●●●   | ○○○○  |
| DINOv2-L | ●○○● | ○●○○ | ●●●● | ●●●● | ●●●●   | ○○○○  |
| DINOv2-G | ●○○○ | ●●●● | ●●●● | ●●●● | ●●●●   | ○○○○  |
| CLIP-B   | ○●○○ | ●●●● | ●●●● | ●●●● | ○○○○   | ○○○○  |
| CLIP-L   | ●●●● | ●●●● | ●●●● | ●●●● | ○○○○   | ○○○○  |
| SigLIP-B | ○○●● | ●●●● | ●●●● | ●●●● | ●●●●   | ●●●●  |

Table 31: **Verdicts under all four null $\times$ statistic constructions, per cell.** Each cell shows four symbols in the order Haar $\times$ p99.9 (the record), Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum; ● = genuine (BH-corrected  $p \leq 0.05$  over the 72 cells of that construction), ○ = not. Genuine counts: 49/72, 46/72, 52/72, 52/72. The Haar construction reproduces the sample spectrum exactly; the Gaussian one does not at small  $n$  and inflates excesses for low-rank clouds (Table 32). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet (genuine under both p99.9 constructions, above every replicate under both supremum constructions); the 99.9th percentile is decided by the bulk.

| model    | IN     |        |          | C100   |        |          | C10    |        |          |
|----------|--------|--------|----------|--------|--------|----------|--------|--------|----------|
|          | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. |
| ViT-T    | -0.005 | -0.014 | -0.038   | -0.044 | -0.031 | -0.055   | -0.057 | -0.027 | -0.050   |
| ViT-S    | -0.023 | -0.025 | -0.052   | -0.034 | -0.025 | -0.049   | -0.098 | -0.063 | -0.090   |
| ViT-B    | -0.027 | -0.025 | -0.054   | -0.018 | -0.011 | -0.039   | -0.090 | -0.054 | -0.078   |
| ViT-L    | -0.028 | -0.029 | -0.040   | -0.047 | -0.034 | -0.051   | -0.104 | -0.066 | -0.096   |
| DINO-B   | -0.005 | -0.002 | -0.008   | -0.068 | -0.056 | -0.068   | -0.100 | -0.070 | -0.097   |
| DINOv2-S | +0.010 | +0.015 | +0.013   | -0.050 | -0.029 | -0.051   | -0.093 | -0.083 | -0.098   |
| DINOv2-B | +0.004 | +0.015 | +0.006   | -0.042 | -0.007 | -0.041   | -0.132 | -0.120 | -0.124   |
| DINOv2-L | -0.013 | +0.000 | -0.011   | -0.053 | -0.008 | -0.044   | -0.153 | -0.136 | -0.122   |
| DINOv2-G | -0.000 | +0.016 | +0.001   | -0.080 | -0.032 | -0.074   | -0.155 | -0.138 | -0.110   |
| CLIP-B   | -0.028 | -0.028 | -0.048   | -0.046 | -0.036 | -0.053   | -0.079 | -0.045 | -0.075   |
| CLIP-L   | -0.016 | -0.017 | -0.029   | -0.042 | -0.031 | -0.052   | -0.087 | -0.049 | -0.082   |
| SigLIP-B | -0.019 | -0.020 | -0.044   | -0.042 | -0.032 | -0.046   | -0.096 | -0.056 | -0.081   |

Table 32: Excess under three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients (the paper’s), Haar-rotated coefficients (exact sample spectrum) and PC-permutation (exact per-component marginals). The Gaussian and PC-permutation constructions agree; the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- $C$  sets (most visibly DINOv2 on CIFAR-100), so the size of those excesses is partly a property of the null hypothesis, while signs and the ImageNet readings are stable.

| model    | IN     |     | C100                |     | C10    |     | DTD    |     | FMNIST              |     | MNIST               |     |
|----------|--------|-----|---------------------|-----|--------|-----|--------|-----|---------------------|-----|---------------------|-----|
|          | exc.   | $r$ | exc.                | $r$ | exc.   | $r$ | exc.   | $r$ | exc.                | $r$ | exc.                | $r$ |
| ViT-T    | -0.012 | 200 | -0.014 <sup>°</sup> | 191 | -0.043 | 198 | -0.017 | 199 | +0.007 <sup>°</sup> | 73  | -0.026 <sup>°</sup> | 167 |
| ViT-S    | -0.017 | 200 | -0.017              | 200 | -0.073 | 200 | -0.021 | 200 | -0.049              | 200 | -0.036 <sup>°</sup> | 183 |
| ViT-B    | -0.022 | 200 | -0.006 <sup>°</sup> | 170 | -0.066 | 200 | -0.021 | 200 | -0.041              | 199 | -0.010 <sup>°</sup> | 130 |
| ViT-L    | -0.021 | 200 | -0.024              | 200 | -0.080 | 200 | -0.024 | 200 | -0.066              | 200 | -0.043              | 196 |
| DINO-B   | -0.021 | 200 | -0.035              | 200 | -0.070 | 200 | -0.024 | 200 | -0.006 <sup>°</sup> | 116 | -0.038 <sup>°</sup> | 193 |
| DINOv2-S | -0.014 | 200 | -0.033              | 200 | -0.084 | 200 | -0.018 | 200 | -0.054              | 200 | -0.004 <sup>°</sup> | 113 |
| DINOv2-B | -0.013 | 200 | -0.032              | 200 | -0.140 | 200 | -0.021 | 200 | -0.081              | 200 | -0.019 <sup>°</sup> | 147 |
| DINOv2-L | -0.010 | 200 | -0.029              | 200 | -0.151 | 200 | -0.024 | 200 | -0.051              | 200 | -0.020 <sup>°</sup> | 145 |
| DINOv2-G | -0.006 | 200 | -0.029              | 200 | -0.169 | 200 | -0.017 | 200 | -0.064              | 200 | -0.023 <sup>°</sup> | 157 |
| CLIP-B   | -0.017 | 200 | -0.023              | 200 | -0.039 | 197 | -0.031 | 200 | -0.038 <sup>°</sup> | 185 | +0.002 <sup>°</sup> | 93  |
| CLIP-L   | -0.017 | 200 | -0.025              | 200 | -0.045 | 199 | -0.031 | 200 | -0.038 <sup>°</sup> | 192 | -0.008 <sup>°</sup> | 124 |
| SigLIP-B | -0.013 | 200 | -0.025              | 200 | -0.053 | 200 | -0.027 | 200 | -0.052              | 199 | -0.044              | 197 |

Table 33: **The angular reading agrees with the Euclidean census.** Centroids L2-normalized, spherical geodesic distances, Haar null built on the normalized cloud and re-normalized, 99.9th-percentile statistic, 200 replicates, BH over 72 cells (°: not genuine); a robustness reading, not the record. Sign-negative 70/72; genuine 56/72, 22/24 on ImageNet+CIFAR-100; verdict agreement with the Euclidean record 65/72 cells. Text (same protocol on the padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance.

| model    | excess (reference) | bootstrap mean | bootstrap s.d. | fraction negative |
|----------|--------------------|----------------|----------------|-------------------|
| ViT-T    | -0.0055            | -0.0051        | 0.0011         | 1.00              |
| ViT-S    | -0.0126            | -0.0122        | 0.0007         | 1.00              |
| ViT-B    | -0.0165            | -0.0160        | 0.0006         | 1.00              |
| ViT-L    | -0.0160            | -0.0160        | 0.0004         | 1.00              |
| DINO-B   | -0.0209            | -0.0208        | 0.0003         | 1.00              |
| DINOv2-S | -0.0164            | -0.0162        | 0.0004         | 1.00              |
| DINOv2-B | -0.0167            | -0.0164        | 0.0002         | 1.00              |
| DINOv2-L | -0.0136            | -0.0134        | 0.0001         | 1.00              |
| DINOv2-G | -0.0080            | -0.0083        | 0.0004         | 1.00              |
| CLIP-B   | -0.0068            | -0.0069        | 0.0009         | 1.00              |
| CLIP-L   | -0.0094            | -0.0095        | 0.0006         | 1.00              |
| SigLIP-B | -0.0080            | -0.0081        | 0.0006         | 1.00              |

Table 34: **The ImageNet excess is stable under resampling of the 100 training images per class.** 30 bootstrap resamples per backbone under the record protocol (Haar null, 99.9th-percentile statistic; 20 null replicates per resample). The bootstrap s.d. of the excess is at most 0.0011; every resample of every backbone is sign-negative.

## A.16 ORC EDGE TYPES

High mean ORC coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges (about twice as often as within superclasses on ImageNet; Table 35), while the mean measures within-cluster compactness. DINOv2 on ImageNet inverts the pattern, consistent with its at-null supremum reading, and recovers it on CIFAR-100.

| model    | dataset | ORC within | ORC across | %neg within | %neg across |
|----------|---------|------------|------------|-------------|-------------|
| ViT-T    | IN      | 0.35       | 0.27       | 5.2         | 9.4         |
| ViT-T    | C100    | 0.51       | 0.42       | 1.4         | 2.7         |
| ViT-S    | IN      | 0.37       | 0.28       | 3.4         | 8.3         |
| ViT-S    | C100    | 0.61       | 0.48       | 0.0         | 3.0         |
| ViT-B    | IN      | 0.40       | 0.30       | 2.1         | 5.9         |
| ViT-B    | C100    | 0.61       | 0.48       | 0.0         | 1.7         |
| ViT-L    | IN      | 0.40       | 0.31       | 2.1         | 6.8         |
| ViT-L    | C100    | 0.61       | 0.47       | 0.6         | 2.1         |
| DINO-B   | IN      | 0.40       | 0.33       | 2.8         | 4.1         |
| DINO-B   | C100    | 0.60       | 0.45       | 0.0         | 1.4         |
| DINOv2-S | IN      | 0.37       | 0.33       | 1.8         | 1.2         |
| DINOv2-S | C100    | 0.60       | 0.44       | 0.0         | 0.8         |
| DINOv2-B | IN      | 0.58       | 0.64       | 0.3         | 0.1         |
| DINOv2-B | C100    | 0.66       | 0.54       | 0.0         | 0.1         |
| DINOv2-L | IN      | 0.69       | 0.72       | 0.0         | 0.0         |
| DINOv2-L | C100    | 0.70       | 0.64       | 0.0         | 0.1         |
| DINOv2-G | IN      | 0.70       | 0.74       | 0.0         | 0.0         |
| DINOv2-G | C100    | 0.77       | 0.70       | 0.0         | 0.0         |
| CLIP-B   | IN      | 0.36       | 0.29       | 4.2         | 7.9         |
| CLIP-B   | C100    | 0.60       | 0.46       | 0.0         | 3.1         |
| CLIP-L   | IN      | 0.35       | 0.27       | 4.0         | 8.4         |
| CLIP-L   | C100    | 0.57       | 0.43       | 0.0         | 2.2         |
| SigLIP-B | IN      | 0.34       | 0.26       | 5.0         | 7.5         |
| SigLIP-B | C100    | 0.59       | 0.43       | 0.0         | 2.3         |

Table 35: Edge-type ORC on the centroid kNN graph (30 WordNet superclasses): negative curvature concentrates on between-cluster bridges.

#### A.17 PROMPT ROBUSTNESS

| model    | photo | name  | image | closeup | this is | wild  | def.  | gloss only | concept | discuss. |
|----------|-------|-------|-------|---------|---------|-------|-------|------------|---------|----------|
| BGE-base | 0.124 | 0.118 | 0.125 | 0.126   | 0.125   | 0.115 | 0.119 | 0.115      | 0.119   | 0.121    |
| E5-base  | 0.114 | 0.119 | 0.117 | 0.120   | 0.115   | 0.116 | 0.108 | 0.115      | 0.120   | 0.116    |
| GPT-2    | 0.131 | 0.145 | 0.110 | 0.086   | 0.080   | 0.089 | 0.105 | 0.124      | 0.112   | 0.105    |
| GPT-2-M  | 0.091 | 0.122 | 0.110 | 0.100   | 0.134   | 0.104 | 0.081 | 0.107      | 0.112   | 0.098    |

Table 36:  $\hat{\delta}$  across prompt templates (ImageNet class names; “gloss only” contains no class name).

| template   | GPT-2 S | GPT-2 M | GPT-2 L | GPT-2 XL |
|------------|---------|---------|---------|----------|
| photo      | −0.016* | −0.019* | −0.043* | −0.050*  |
| name_only  | −0.006  | −0.007  | −0.048* | −0.051*  |
| image      | −0.018* | −0.028* | −0.050* | −0.049*  |
| closeup    | +0.007  | −0.018* | −0.044* | −0.049*  |
| this_is    | −0.032* | −0.022* | −0.042* | −0.046*  |
| wild       | −0.023* | −0.067* | −0.014  | −0.010*  |
| def_pair   | −0.002  | −0.019* | −0.036* | −0.035*  |
| gloss_only | −0.020* | −0.030* | −0.033* | −0.036*  |
| concept    | −0.002  | −0.003  | −0.043* | −0.041*  |
| discussion | −0.010  | −0.013* | −0.041* | −0.040*  |

Table 37: Template-conditioned excess over the spectrum-matched null for the four GPT-2 sizes (10 templates; \*:  $|z| \geq 2$  against combined null and estimator noise). The scale trend of §4 is assessed on every template, not only on the paper’s baseline. Padding-free extraction (one prompt at a time, the canonical protocol of Table 12). Mean excess over templates: −0.012 (S), −0.023 (M), −0.040 (L), −0.041 (XL); cells at  $|z| \geq 2$ : 5, 8, 9 and 10 of 10. Only 3 null replicates per cell here, so the  $z$  are coarser than the census’s 20-replicate values: on the baseline template the 200-replicate census (Table 12) finds S and M not genuine (excess −0.013/−0.009,  $p = 0.09/0.14$ ), so their starred cells are borderline, whereas the L/XL cells ( $z$  from −4 to −9) are not in doubt. Template sensitivity of raw  $\hat{\delta}$ : range up to 0.065 for causal LMs and  $\leq 0.012$  for embedders (prompt table in Appendix 36).

#### A.18 SAMPLE-LEVEL FEATURES

| model    | CIFAR-100 (10 images/class, $n=1000$ ) |                       |          |                 | DTD (22 images/class, $n=1034$ ) |                       |          |                 |
|----------|----------------------------------------|-----------------------|----------|-----------------|----------------------------------|-----------------------|----------|-----------------|
|          | sup. $\hat{\delta}$                    | $\hat{\delta}_{99.9}$ | excess   | $r/200$ ( $p$ ) | sup. $\hat{\delta}$              | $\hat{\delta}_{99.9}$ | excess   | $r/200$ ( $p$ ) |
| ViT-T    | 0.130                                  | 0.093                 | −0.0016° | 134 (0.333)     | 0.124                            | 0.087                 | −0.0004° | 111 (0.448)     |
| ViT-S    | 0.121                                  | 0.086                 | +0.0051° | 21 (0.896)      | 0.103                            | 0.070                 | −0.0062  | 197 (0.020)     |
| ViT-B    | 0.121                                  | 0.082                 | +0.0031° | 50 (0.751)      | 0.092                            | 0.060                 | −0.0080  | 200 (0.005)     |
| ViT-L    | 0.100                                  | 0.067                 | −0.0041° | 185 (0.080)     | 0.092                            | 0.056                 | −0.0073  | 200 (0.005)     |
| DINO-B   | 0.104                                  | 0.070                 | −0.0042° | 194 (0.035)     | 0.110                            | 0.075                 | +0.0008° | 80 (0.602)      |
| DINOv2-S | 0.100                                  | 0.065                 | −0.0049  | 200 (0.005)     | 0.145                            | 0.090                 | +0.0058° | 0 (1.000)       |
| DINOv2-B | 0.089                                  | 0.050                 | −0.0090  | 200 (0.005)     | 0.139                            | 0.073                 | −0.0019° | 162 (0.194)     |
| DINOv2-L | 0.087                                  | 0.043                 | −0.0131  | 200 (0.005)     | 0.131                            | 0.064                 | −0.0060  | 200 (0.005)     |
| DINOv2-G | 0.102                                  | 0.036                 | −0.0232  | 200 (0.005)     | 0.136                            | 0.065                 | −0.0079  | 200 (0.005)     |
| CLIP-B   | 0.117                                  | 0.080                 | −0.0060° | 184 (0.085)     | 0.125                            | 0.089                 | +0.0021° | 71 (0.647)      |
| CLIP-L   | 0.132                                  | 0.084                 | +0.0037° | 35 (0.826)      | 0.120                            | 0.082                 | −0.0008° | 124 (0.383)     |
| SigLIP-B | 0.113                                  | 0.078                 | −0.0074  | 199 (0.010)     | 0.114                            | 0.079                 | −0.0044° | 184 (0.085)     |

Table 38: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on  $\approx 1000$  stratified training images per cell (the per-image features, not class centroids): sup.  $\hat{\delta}$  is the raw supremum  $\delta_{\text{norm}}$  the literature reports;  $\hat{\delta}_{99.9}$ , excess,  $r/200$  and  $p$  follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); °: not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most −0.023.

## A.19 TEXT-MODALITY TRANSFER

| model    | supremum, Gaussian null |        | census of record      |        |                 | trip. (cos) | NC H-R (pp) | FS H-R (pp) |
|----------|-------------------------|--------|-----------------------|--------|-----------------|-------------|-------------|-------------|
|          | $\hat{\delta}_{\max}$   | excess | $\hat{\delta}_{99.9}$ | excess | $r/200$ ( $p$ ) |             |             |             |
| BGE-base | 0.122                   | -0.028 | 0.082                 | -0.021 | 200 (0.005)     | 0.89        | -0.59       | +0.05±0.05  |
| E5-base  | 0.127                   | -0.024 | 0.086                 | -0.017 | 200 (0.005)     | 0.89        | -0.43       | +0.04±0.03  |
| GTE-base | 0.132                   | -0.020 | 0.088                 | -0.017 | 200 (0.005)     | 0.89        | -0.82       | +0.08±0.06  |

Table 39: DBpedia Classes (219 leaf classes, 3 levels): excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates;  $r$  = replicates above the real value, left-tail  $p$ ), strong sibling alignment, and  $\hat{\delta}$  above the low- $\delta$  band correctly predicting marginal gains. Tree map on the same classes: no configuration degenerates (largest cluster  $\leq 10\%$ ), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82.

## A.20 HIERARCAPS MULTI-LEVEL HIERARCHY

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 40).

| model    | $\rho(\text{level, radius})$ | % monotone | trip. L1 | trip. L2 | $\hat{\delta}$ leaves | excess |
|----------|------------------------------|------------|----------|----------|-----------------------|--------|
| BGE-base | +0.80±0.31                   | 49.6       | 0.88     | 0.97     | 0.130                 | +0.003 |
| E5-base  | +0.87±0.19                   | 53.9       | 0.90     | 0.97     | 0.139                 | +0.010 |
| GTE-base | +0.47±0.55                   | 24.0       | 0.90     | 0.97     | 0.135                 | +0.006 |
| CLIP-B   | +0.64±0.43                   | 27.2       | 0.86     | 0.95     | 0.123                 | -0.011 |
| CLIP-L   | +0.51±0.49                   | 18.7       | 0.86     | 0.96     | 0.117                 | -0.010 |
| SigLIP-B | +0.32±0.55                   | 9.8        | 0.83     | 0.91     | 0.138                 | +0.014 |

Table 40: HierarCaps test set (1000 four-level chains): radial ordering under the paper’s projection (chance for strict monotonicity =  $1/24 \approx 4.2\%$ ; shuffle controls  $\approx 0$ ), sibling triplet agreement (cosine; chance 0.5), and leaf  $\hat{\delta}$  with spectrum-null excess.

A.21 PARALLELOGRAM DEFECT  $\xi$ : A NORM-STRUCTURE DESCRIPTOR

**$\xi$  reads norm structure, not curvature.** A parallelogram estimator  $\xi$ , intended as a curvature sign, is driven negative by heterogeneous norms alone, so it is reported only as a norm-structure descriptor. For a triangle  $(a, b, c)$  with  $m$  the approximate midpoint of  $(b, c)$  (the minimizer of  $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$ ), the parallelogram defect is  $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2 D(a, m) \text{diam})$ , averaged over  $2 \times 10^4$  sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

| space / model                   | $\xi$ (mean) | frac. negative | null $\xi$         | excess ( $\pm$ CI95) | $z$ |
|---------------------------------|--------------|----------------|--------------------|----------------------|-----|
| iid Gaussian ( $d=768$ )        | +0.092       | 0%             | —                  | —                    | —   |
| $S^d$ (geodesic)                | +0.113       | 0%             | —                  | —                    | —   |
| $S^d$ (chord)                   | +0.115       | 0%             | —                  | —                    | —   |
| $\mathbb{H}^2$ region ( $R=6$ ) | -0.194       | 96%            | —                  | —                    | —   |
| balanced tree (depth 9)         | -0.239       | 96%            | —                  | —                    | —   |
| ViT-T                           | +0.027       | 25%            | +0.027 $\pm$ 0.002 | -0.001 $\pm$ 0.003   | -0  |
| ViT-S                           | +0.027       | 23%            | +0.042 $\pm$ 0.002 | -0.015 $\pm$ 0.005   | -7  |
| ViT-B                           | +0.033       | 16%            | +0.047 $\pm$ 0.002 | -0.014 $\pm$ 0.004   | -6  |
| ViT-L                           | +0.028       | 21%            | +0.046 $\pm$ 0.002 | -0.018 $\pm$ 0.005   | -7  |
| DINO-B                          | -0.008       | 55%            | +0.052 $\pm$ 0.003 | -0.059 $\pm$ 0.005   | -21 |
| DINOv2-S                        | -0.020       | 66%            | +0.061 $\pm$ 0.004 | -0.081 $\pm$ 0.008   | -20 |
| DINOv2-B                        | -0.049       | 83%            | +0.075 $\pm$ 0.003 | -0.123 $\pm$ 0.005   | -45 |
| DINOv2-L                        | -0.068       | 90%            | +0.080 $\pm$ 0.002 | -0.149 $\pm$ 0.004   | -66 |
| DINOv2-G                        | -0.070       | 89%            | +0.082 $\pm$ 0.002 | -0.152 $\pm$ 0.004   | -69 |
| CLIP-B                          | +0.019       | 32%            | +0.024 $\pm$ 0.002 | -0.005 $\pm$ 0.003   | -3  |
| CLIP-L                          | +0.028       | 28%            | +0.032 $\pm$ 0.002 | -0.004 $\pm$ 0.004   | -2  |
| SigLIP-B                        | +0.036       | 20%            | +0.031 $\pm$ 0.002 | +0.005 $\pm$ 0.004   | +2  |

Table 41: Parallelogram curvature-sign estimator  $\xi$  on reference geometries (top) and ImageNet centroids (bottom), with the spectrum-matched null, excess and  $z$ -score for each model (20 null replicates; noise combines null and estimator s.d.): matched nulls have positive  $\xi$ , so raw readings understate curvature.

| model    | $\xi_{\text{geo}}$ | frac. neg. | excess  | $z$   |
|----------|--------------------|------------|---------|-------|
| ViT-T    | +0.031             | 0.24       | -0.0048 | -2.7  |
| ViT-S    | +0.065             | 0.08       | +0.0092 | +6.2  |
| ViT-B    | +0.067             | 0.07       | +0.0144 | +10.5 |
| ViT-L    | +0.074             | 0.04       | +0.0113 | +7.9  |
| DINO-B   | +0.085             | 0.05       | +0.0014 | +1.0  |
| DINOv2-S | +0.100             | 0.02       | +0.0033 | +2.3  |
| DINOv2-B | +0.108             | 0.01       | +0.0046 | +4.9  |
| DINOv2-L | +0.108             | 0.01       | +0.0040 | +3.9  |
| DINOv2-G | +0.094             | 0.01       | -0.0114 | -12.5 |
| CLIP-B   | +0.025             | 0.28       | -0.0129 | -7.1  |
| CLIP-L   | +0.033             | 0.25       | -0.0120 | -4.8  |
| SigLIP-B | +0.051             | 0.14       | +0.0025 | +1.2  |

Table 42:  $\xi$  on the angular geometry: centroids L2-normalized, distances the spherical geodesic, spectrum null built on the normalized cloud and re-normalized (20 replicates; ImageNet centroid store). Probes whether the deep negative  $\xi$ -excess of DINOv2 on ImageNet (Table 41) is a norm-structure effect. Outcome: it is for S/B/L, whose angular  $\xi$ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere: the Euclidean  $\xi$  signal of DINOv2 is largely carried by norm structure.

| reference (no curvature unless stated)     | $\xi$   | frac. negative |
|--------------------------------------------|---------|----------------|
| iid Gaussian                               | +0.0919 | 0.00           |
| Gaussian, lognormal radii ( $\sigma=0.3$ ) | -0.0621 | 0.99           |
| Gaussian, lognormal radii ( $\sigma=0.6$ ) | -0.0607 | 1.00           |
| Gaussian, lognormal radii ( $\sigma=1.0$ ) | -0.0496 | 0.99           |
| core + shell mixture                       | -0.3217 | 0.79           |
| uniform sphere                             | +0.1154 | 0.00           |
| sphere with radial jitter                  | -0.0571 | 1.00           |

Table 43:  $\xi$  on references with heterogeneous norms but no curvature ( $n=1000$ ,  $d=768$ , 5 seeds). Norm dispersion alone drives  $\xi$  negative (lognormal radii, radial jitter on a sphere, core+shell): the estimator reads norm structure, not curvature, and the paper reports it only as such.

## B ILLUSTRATIVE VISUALIZATION

Figure 11 is a low-dimensional illustration of the hub-and-spoke organization; every quantitative comparison in the paper is made in the full feature space (§4).

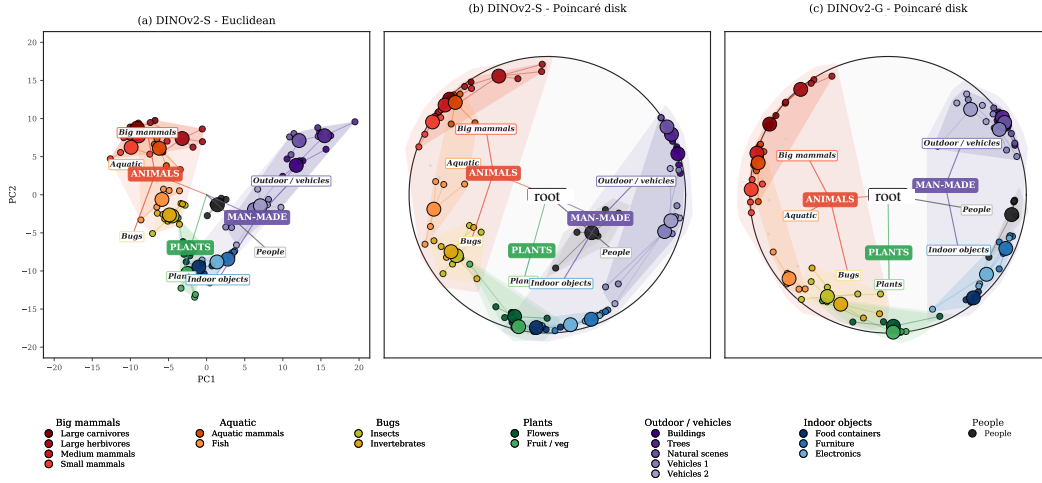


Figure 11: Illustrative 2-D projection of the 100 CIFAR-100 class centroids (per-panel PCA fit on the 20 super-class hubs; panels preserve 22–30% of hub variance and 6–10% of spoke length, so visual compactness is not comparable across panels and quantitative comparisons are made in full dimension). (a) DINOv2-S, Euclidean; (b) DINOv2-S after the projection of §3; (c) DINOv2-G.